Coupled Temporal Variation Information Estimation and Resolution Enhancement for Remote Sensing Spatial–Temporal–Spectral Fusion

Weiwei Sun, *Senior Member, IEEE*, Jun Zhou, Xiangchao Meng, *Member, IEEE*, Gang Yang, *Member, IEEE*, Kai Ren, and Jiangtao Peng, *Senior Member, IEEE*

***Abstract*— Spatial–temporal–spectral fusion (STSF) of remote sensing imagery can produce data with the highest spatial and spectral resolution, only as well as fine temporal resolution, by integrating images with complementary information in both the temporal and spectral domains. Accuracy of temporal variation is an important guarantee for achieving fidelity fusion in STSF. However, current STSF methods estimate the temporal variation only by utilizing the temporal variation between the observed multispectral image (MSI) and the relationship between MSI and hyperspectral image (HSI), which is difficult to obtain accurate temporal variation. To address this problem, this article proposes a coupled temporal variation information estimation and resolution enhancement for remote sensing image STSF (CTVRE-STSF). The temporal variation information estimation model estimates the temporal variation of the target image, while the resolution enhancement model provides additional constraints for estimating the temporal variation. For the temporal variation information reconstruction model, we build a temporal variation information estimation based on a generalized linear mixed model and use the temporal variation between MSIs. In addition, a resolution enhancement model is constructed to estimate the temporal variation of the target image by incorporating relevant prior knowledge. The introduction of the resolution enhancement model in the prior provides additional constraints on the estimation of the temporal variation high-dimensional information, thus facilitating the resolution improvement. The experimental results on two real datasets demonstrate the effectiveness and superiority of our proposed method over current state-of-the-art methods, especially in terms of spectral fidelity.**

***Index Terms*— Hyperspectral (HS), image fusion, multispectral (MS), spatial resolution, spectral resolution, temporal resolution.**

Manuscript received 27 August 2023; revised 31 October 2023; accepted 16 November 2023. Date of publication 28 November 2023; date of current version 4 December 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 42122009, Grant 41971296, Grant 42171326, Grant 41801252, Grant 41801256, and Grant 61871177; in part by the Zhejiang Provincial Natural Science Foundation of China under Grant LR19D010001, Grant LQ18D010001, and Grant LY22F010014; in part by the Public Projects of Ningbo City under Grant 2021S089; and in part by the Graduate Student Scientific Research and Innovation Fund of Ningbo University under Grant Y202249916. *(Corresponding author: Weiwei Sun.)*

Weiwei Sun, Jun Zhou, and Gang Yang are with the Department of Geography and Spatial Information Techniques, Ningbo University, Ningbo 315211, China (e-mail: sunweiwei@nbu.edu.cn; junzhounbu1@gmail.com; yanggang@nbu.edu.cn).

Xiangchao Meng and Kai Ren are with the Faculty of Electrical Engineering and Computer Science, Ningbo University, Ningbo 315211, China (e-mail: mengxiangchao@nbu.edu.cn; 1811073013@nbu.edu.cn).

Jiangtao Peng is with the Hubei Key Laboratory of Applied Mathematics, Faculty of Mathematics and Statistics, Hubei University, Wuhan 430062, China (e-mail: pengjt1982@hubu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3335418

I. Introduction

SATELLITE remote sensing image with high spatial, spectral, and fine temporal resolution has garnered considerable attention from researchers. High spatial and fine temporal resolution images can provide powerful support for monitoring rapidly changing surface cover, such as coastal wetlands and crop areas [1], [2], [3], [4], [5], [6]. However, due to technical and cost limitations, satellite sensors have to make a fundamental tradeoff between temporal, spatial, and spectral resolutions. No single satellite can provide finely spatial global coverage of high spectral resolution data with dense time series. Different satellite sensors typically have complementary resolution characteristics. Hyperspectral (HS) sensors can provide data on hundreds of spectral bands, enabling fine identification of land features [7]. Their relatively coarse temporal and low spatial resolutions limit their application in rapidly changing and complex areas. On the other hand, multispectral (MS) sensors typically have relatively higher spatial and finer temporal resolutions. Directly improving the spatial–temporal resolution of HS sensors or the spectral resolution of MS sensors is very difficult and costly. Therefore, a spatial–temporal–spectral fusion (STSF) method that can integrate complementary information from existing multisource remote sensing images is beneficial. In this article, we propose an STSF method that is tailored to the characteristics of existing satellite data, combining low spatial and coarse temporal resolution HS image (HSI) with high spatial and fine temporal resolution MS image (MSI) to generate the desired images with the highest spatial, spectral, and fine temporal resolution.

A. Related Studies

Remote sensing image fusion is an effective method to fuse complementary information between multiple source observations [8], [9], [10]. So far, many fusion algorithms have been proposed. According to the classification in the literature [11], existing fusion methods are categorized into spatial–spectral fusion (SSF), spatial–temporal fusion (STF), and STSF.

SSF aims to obtain images with high spatial and spectral resolutions by fusing images with high spatial but low spectral resolutions and images with high spectral but low spatial resolutions [12], [13], [14], [15], [16], [17]. Numerous SSF

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:25:27 UTC from IEEE Xplore. Restrictions apply.

methods have been proposed, which can be generally classified into four categories: pansharpening-based methods [18], matrix factorization-based methods, tensor factorization-based methods, and deep learning-based methods. Pansharpening-based methods are the earliest SSF methods, which mainly include component substitution (CS) methods and multiresolution analysis (MRA) methods. CS methods first project the upsampled low spatial resolution image into a new space and then replace the spatial components representing spatial information with those from the high spatial resolution image [19], [20], [21]. In MRA methods, the fusion results are obtained by injecting the low-frequency information of the high spatial resolution image obtained from multiresolution decomposition into the low spatial resolution image [22], [23], [24]. Pansharpening-based methods are relatively simple but may not establish an explicit relationship model, resulting in spatial or spectral distortions [25]. Matrix factorization-based methods are often based on the assumption of spectral unmixing or sparse representation theory, which assumes that the target fused image along the spectral dimension can be decomposed into a spectral basis containing spectral information multiplied by a coefficient matrix containing spatial information [26]. Some prior information about the target image is also introduced to aid the decomposition, including nonnegative constraints [27], sparse constraints [28], [29], [30], total variation constraints [31], [32], and low-rank constraints [33], [34], [35]. These methods are mathematically rigorous and provide more accurate results. Tensor decomposition-based methods directly decompose 3-D HSI, effectively preserving structural information [30], [36], [37], [38], [39], [40], resulting in more accurate fusion performance [41]. Recently, deep learning-based SSF methods have emerged [42], [43], [44], [45], which exploit the feature extraction and learning capabilities of deep neural networks. These methods train a network model to learn the mapping between observed and ideal images, but they often require additional training data and may lack interpretability, limiting their generalizability [46].

STF aims to obtain images with both spatial resolution and fine temporal by integrating images with high spatial resolution but coarse temporal resolution with images with low spatial resolution but fine temporal resolution [9], [47], [48], [49], [50]. STF methods can be divided into three categories: weight function-based, spatial unmixing-based, and learning-based. The spatial and temporal adaptive reflectance fusion model (STARFM) is a pioneering work in weight function-based methods [51], which searches for neighboring similar pixels in the observed low spatial resolution image and represents the center pixel of the predicted temporal image based on a designed weighting function. Several derivative methods have been proposed, including enhanced spatial and temporal adaptive reflectance fusion model (ESTARFM) [52] and spatial temporal adaptive algorithm for mapping reflectance change (STAARCH) [53]. Spatial unmixing-based methods are based on a fine image (or fine classification map) and estimate the unknown class abundance based on the known class proportion of coarse pixels provided by the fine classification map [54], [55]. It assumes no change in land cover during the predicted period, and the class proportion of low-resolution images remains constant over time [56]. Learning-based methods commonly utilize machine learning algorithms, such as dictionary learning [57], [58] and deep neural networks [59], [60], [61], [62] to extract temporal variation from pairs of low spatial resolution images and incorporate temporal variation information into high spatial resolution images based on the relationship between low and high temporal resolution images.

STSF fuses all three attributes simultaneously, resulting in high spatial and spectral resolution images with fine temporal resolution [11]. To date, some STSF methods have been proposed, which can be divided into step-by-step STSF and integrated STSF based on the fusion process. The step-by-step STSF methods perform fusion in two or three steps, including SSF, STF, and temporal–spectral fusion (TSF). For instance, a spatial–temporal–spectral blending model using satellite images first performs the STF of MODIS-Landsat Thematic Mapper (TM) to increase the spatial resolution of MODIS to TM image resolution and then conducts the TSF of the resolution-enhanced fusion image and Hyperion HSI [63]. Zhao and Huang [64] developed a hybrid image fusion model based on Landsat 8 Operational Land Imager (OLI)-MODIS-Hyperion images, including SSF of OLI MS and PAN images with the Hyperion HSI, STF of MODIS and OLI, and TSF of the high-resolution Hyperion HSI and the resolution-enhanced MODIS image. Tang et al. [65] separately performed STF and SSF of Sentinel-2 and Sentinel-3 images to obtain high spatial–temporal–spectral resolution Sentinel-3 images. Although these methods simplify the fusion problem, they introduce error accumulation and additional time consumption.

In contrast to step-by-step STSF methods, integrated STSF methods only require a single fusion process to obtain the fusion image. Shen [66] constructed the first integrated STSF framework and conducted experiments on a simulated dataset. In their later work, they further proposed a multisensor integrated STSF mode [11]. Huang et al. [67] proposed a unified STSF method based on Bayesian framework. Zhao et al. [68] proposed an integrated STSF method based on sparse representation theory, utilizing the similarity of ideal images in the spectral dimension and local similarity in the spatial dimension. Integrated STSF based on the tensor perspective has also been explored. Peng et al. [69] proposed a sparse tensor decomposition STSF method based on the assumption that the target fusion image is sparse in temporal, spatial, and spectral dimensions. Li et al. [70] proposed an anisotropic sparsity-constrained low-rank tensor approximation STSF by using an anisotropic sparsity tensor norm to explore the higher order structure of the fused images. Deep learning-based STSF methods have recently gained prominence. Liu et al. [10] introduced an unsupervised deep learning-based STSF fusion model that explores spatial–temporal variations and spectral mapping relationship between MSI and HSI. The model uses two networks and a spatial–spectral interaction compensation module to produce a final fused image.

Although a few integrated STSF methods have been developed, current STSF still faces challenges in adapting to data characteristics and preserving spectral fidelity in the fused

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:25:27 UTC from IEEE Xplore. Restrictions apply.

images. The current integrated STSF methods are primarily designed for MODIS-Landsat with the assumption of a finer temporal resolution of HSI (i.e., MODIS). Unfortunately, the temporal resolution of current HSI is generally coarser compared to MSI. For example, the temporal resolutions of ZY1-02D and PRISMA HSI are 55 and 29 days, respectively. In contrast, some widely used MSI with higher spatial resolution exhibit relatively finer temporal resolution. Sentinel-2 provides observations of four bands with a 10-m spatial resolution every five days, offering a relatively finer temporal resolution than his, thereby limiting the applicability of these methods to the fusion of low spatial and fine temporal resolution HSI. For example, the methods proposed by Shen [66], Peng et al. [69], and Li et al. [70] rely on having HSI at the target time to establish spectral fidelity. However, in the task of STSF for low spatial and coarse temporal resolution HSI, obtaining HSI at the target time is not feasible. In our previous work, generalized linear spectral mixing model for STSF (GLMM-STSF) [71], we proposed an STSF suitable for HSI with low spatial resolution and coarse temporal resolution, taking into account the resolution characteristics of current HSI data for the first time. However, the lack of HSI at the time of prediction results in the absence of the high spectral resolution temporal variation information of the target image, making it difficult to achieve spectral fidelity of the fused image. Our previous work [71] used a generalized linear spectral mixing model to characterize temporal variation and construct constraints using only temporal variation between MSIs. The work of Liu et al. [10] takes time variation information from MSI image pairs and uses spatial and spectral attention modules to correct for the time variation. These methods do not take into full consideration the recovery of high spectral resolution temporal variation information of the target image.

B. Our Contribution

To address the above issues, this article proposes a novel STSF method called the coupled temporal variation information estimation and resolution enhancement for remote sensing image STSF method (CTVRE-STSF), which aims to generate high spatial–spectral resolution image (HR-HSI) at target time by integrating low spatial and coarse temporal resolution HSI and high spatial and fine temporal resolution MSI. First, the HR-HSI at the target time is decomposed into endmember and abundance matrices. The abundance matrix is primarily inferred from the MSI at the target time. The endmember matrix is obtained by element-wise multiplication with the endmember matrix of the observed HSI during acquisition. To estimate temporal variation in endmembers, we leverage the temporal differences between the observed MSIs and the relationship between the observed and target images. Furthermore, we account for the nonlocal similarity and low-rank properties of temporal variation in the target image. This is achieved through the introduction of nonlocal similarity and low tensor multirank priors as regularization terms, optimizing temporal variation information. Ultimately, the estimated temporal variation information of the target image is utilized for estimating the temporal variation of the endmembers.

The main characteristics and contributions of this proposed method can be summarized as follows.

1) A novel STSF approach that combines time variation information estimation with resolution enhancement modeling, making it particularly suitable for fusing low temporal and spatial resolution HSI data, such as PRISMA and ZY1-02D HSI.
2) Addressing the challenge of fusing spectral fidelity in target image by constructing a temporal variation information estimation model and a resolution enhancement model. We employ the GLMM to capture temporal variation in MSI, develop a temporal variation information estimation model for spectral downscaling, and introduce a temporal variation resolution enhancement model guided by prior temporal variation information. Coupling these models results in precise temporal variation estimation, yielding high-fidelity STSF performance.

The organization of this article is as follows. Section II presents the observed model and introduces the time variation model based on the GLMM and the time variation information model. The proposed integrated STSF model is presented in Section III. Experimental results on two real remote sensing datasets are presented in Section IV, and conclusions are given in Section V.

II. Observed Model and Time Variation Model

A. Observed Model

Let $\mathbf{X}_1$ and $\mathbf{X}_2 \in \mathbb{R}^{L\times N}$ with L bands and N pixels denote the high spatial–spectral resolution HR-HSI at observation time $\boldsymbol{t}_1$ and prediction time $\boldsymbol{t}_2$, respectively. $\mathbf{Y}_1$ and $\mathbf{Y}_2 \in \mathbb{R}^{L\times M}$, with L bands and M pixels and $M < N$, represent the high spectral but low spatial and coarse temporal resolution HSI at $\boldsymbol{t}_1$ and $\boldsymbol{t}_2$, respectively. $\mathbf{Z}_1$ and $\mathbf{Z}_2 \in \mathbb{R}^{l\times N}$, with l bands and N pixels ($l < L$), denote the high spatial and fine temporal but low spectral resolution MSI at $\boldsymbol{t}_1$ and $\boldsymbol{t}_2$, respectively.

A linear spectral mixing model (LMM) is widely used in fusion problems because of its simplicity and effectiveness. The LMM assumes that each pixel in the observed image can be modeled as a convex combination of the endmember's spectral present in the scene. Formally, the HR-HSI $\mathbf{X}_i$ can be expressed as

$$\mathbf{X}_i = \mathbf{E}_i\mathbf{A}_i + \boldsymbol{\varepsilon}_i \tag{1}$$

where $\mathbf{X}_i \in \mathbb{R}^{L\times N}$, $i \in \{1, 2\}$, denotes the highest spatial, spectral, and fine temporal resolution images at $\boldsymbol{t}_i$. Matrices $\mathbf{E}_i \in \mathbb{R}^{L\times Q}$ and $\mathbf{A}_i \in \mathbb{R}^{Q\times N}$ denote the endmember matrix and the abundance matrix corresponding to HR-HSI $\mathbf{X}_i$, respectively, Q is the number of endmembers, with $Q \ll L$, and $\boldsymbol{\varepsilon}_i$ is the residual matrix containing the noise and model error.

Let $Y_i \in \mathbb{R}^{L\times M}$ and $Z_i \in \mathbb{R}^{l\times N}$, $i \in \{1, 2\}$, denote the high spectral resolution but low spatial and coarse temporal resolution HSI and high spatial and fine temporal resolution but low spectral resolution MSI observed at $\boldsymbol{t}_i$, respectively. Both HSI Y_i and MSI Z_i can be modeled as degraded versions

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:25:27 UTC from IEEE Xplore. Restrictions apply.

of HR-HSI $\mathbf{X}_i$, i.e.,

$$\mathbf{Y}_i = \mathbf{X}_i\mathbf{FS} \tag{2}$$

$$\mathbf{Z}_i = \mathbf{RX}_i \tag{3}$$

where matrices note the blurring matrix and spatial downsampling matrix, respectively, and matrix $\mathbf{R} \in \mathbb{R}^{l\times L}$ denotes the spectral response function of MSI Z_i. Using (1) and (2), (3) can be further rewritten as

$$\mathbf{Y}_i = \mathbf{E}_i\mathbf{A}_i\mathbf{FS} \tag{4}$$

$$\mathbf{Z}_i = \mathbf{RE}_i\mathbf{A}_i. \tag{5}$$

In this article, the fusion problem is to predict the high spatial and spectral resolutions HR-HSI at $\boldsymbol{t}_2$ based on the MSI and HSI observed at $\boldsymbol{t}_1$ and the MSI observed at $\boldsymbol{t}_2$.

B. GLMM for Spatial–Temporal–Spectral Fusion

The generalized linear mixing model (GLMM) was proposed in [72] with the assumption of variability of spectral endmembers. For each endmember, a band-dependent scale factor is used to model the variation of the spectral endmembers, allowing the model to accommodate arbitrary variations in the endmembers' spectral. GLMM was applied in work [71], the endmembers of the image are considered as a variable, and for each band, the endmember is allowed to vary according to a scale factor to model the spectral variation between HR-HSI at the observation time $\boldsymbol{t}_1$ and the prediction time $\boldsymbol{t}_2$, denoted as

$$\mathbf{E}_2 = \mathbf{\Phi} \odot \mathbf{E}_1 \tag{6}$$

where $\mathbf{\Phi} \in \mathbb{R}^{L\times Q}$ is a matrix of positive scale factors acting on the endmember-wise, $\mathbf{E}_1$ and $\mathbf{E}_2 \in \mathbb{R}^{L\times Q}$ are the corresponding endmember matrices of $\mathbf{X}_1$ and $\mathbf{X}_2$, respectively, and the symbol $\odot$ denotes the Hadamard product. The temporal variation relationship between HR-HSI at two different times can be written as

$$\mathbf{X}_2 = \mathbf{\Phi} \odot \mathbf{E}_1\mathbf{A}_2. \tag{7}$$

The scale factor $\mathbf{\Phi}$ is used to characterize the variation of the spectral between $\boldsymbol{t}_1$ and $\boldsymbol{t}_2$, which allows using the endmembers $\mathbf{E}_1$ of $\mathbf{X}_1$ to obtain the endmembers $\mathbf{E}_2$ of $\mathbf{X}_2$ by estimating the variation $\mathbf{\Phi}$ of the spectral.

C. Time Variation Information Model

In this fusion problem, the absence of HSI $\mathbf{Y}_2$, which holds the key high spectral resolution information of the desired fusion image $\mathbf{X}_2$, causes great difficulty and uncertainty in estimating the spectral variation, which further leads to the absence of spectral information of the fused image $\mathbf{X}_2$, since only MSI $\mathbf{Z}_2$ is available at the prediction time $\boldsymbol{t}_2$. On this basis, we constructed the time variation information model as

$$\mathbf{X}_2 - \mathbf{X}_1 = \mathbf{T} \tag{8}$$

where $\mathbf{T} \in \mathbb{R}^{L\times N}$ is the time variation information. Notice that since $\mathbf{X}_1$ and $\mathbf{X}_2$ are not directly observable in the real scene, $\mathbf{T}$ is a virtual time variation information. By substituting (1) and (7) into (8), we can obtain

$$\mathbf{T} = \mathbf{P} \odot \mathbf{E}_1\mathbf{A}_2 - \mathbf{E}_1\mathbf{A}_1. \tag{9}$$

In our fusion model, $\mathbf{Z}_1$ and $\mathbf{Z}_2$ MSIs pair can be provided simultaneously, and we can give MS differential image $\mathbf{D}$ that retains the temporal variation between the two MSIs

$$\mathbf{Z}_1 - \mathbf{Z}_2 = \mathbf{D}. \tag{10}$$

Substituting into (5) and (6), (10) can be rewritten as

$$\mathbf{D} = \mathbf{R}(\mathbf{P} \odot \mathbf{E}_1\mathbf{A}_2 - \mathbf{E}_1\mathbf{A}_1). \tag{11}$$

Since the MSI can be obtained by spectral degradation of the corresponding HR-HSI, the differential image $\mathbf{D}$ between $\mathbf{Z}_1$ and $\mathbf{Z}_2$ of the MSIs pair should similarly satisfy the spectral degradation constraint of the temporal variation information T

$$\mathbf{D} = \mathbf{RT}. \tag{12}$$

III. CTVRE-STSF

We propose a novel STSF method integrating temporal variation information estimation and resolution enhancement models. In the temporal variation information estimation model (Fig. 1, left), we utilize GLMM and MSI differential images as the basis for constructing the temporal variation information estimation model. This model enables us to estimate the temporal variation of HR-HSI under the spectral downscaling constraint of MSI differential image. In addition, using GLMM, we represent the full-scale temporal variation information of HR-HSI and introduce additional prior information to enhance the high-dimensional information of the temporal variation information (Fig. 1, right). By combining these models, we propose the STSF model, as shown in Fig. 1. Finally, we design algorithms based on alternating least squares (ALS) and alternating directional multiplier method (ADMM) to solve our proposed model effectively.

In the STSF for low spatial and coarse temporal resolution HSI, the absence of HSI during the prediction time hinders the availability of high spectral resolution information for the target image. The spectral information of the target image is in a high-dimensional space, and it is difficult to recover the spectral information with high spectral resolution directly from the MSI. Since the high-dimensional spectral information is preserved in the HSI in the observed time, the spectral information of the target image can be estimated by establishing the temporal variation relationship of the spectra between the two time periods. The temporal variation is considered a perturbation acting on the endmember-by-endmember of each band by Imbiriba et al. [72]. Borsoi et al. [73] used the parametric perturbation model to model temporal variations in the fusion task. In works [70] and [71], it is further verified that temporal variation can be carefully modeled using parametric perturbation models. However, previous studies [10], [71] have only used MSI pairs to estimate temporal variation. Relying only on the temporal variation information of the MSI time difference images is insufficient to fully estimate the temporal variation of the spectral information of the target

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:25:27 UTC from IEEE Xplore. Restrictions apply.

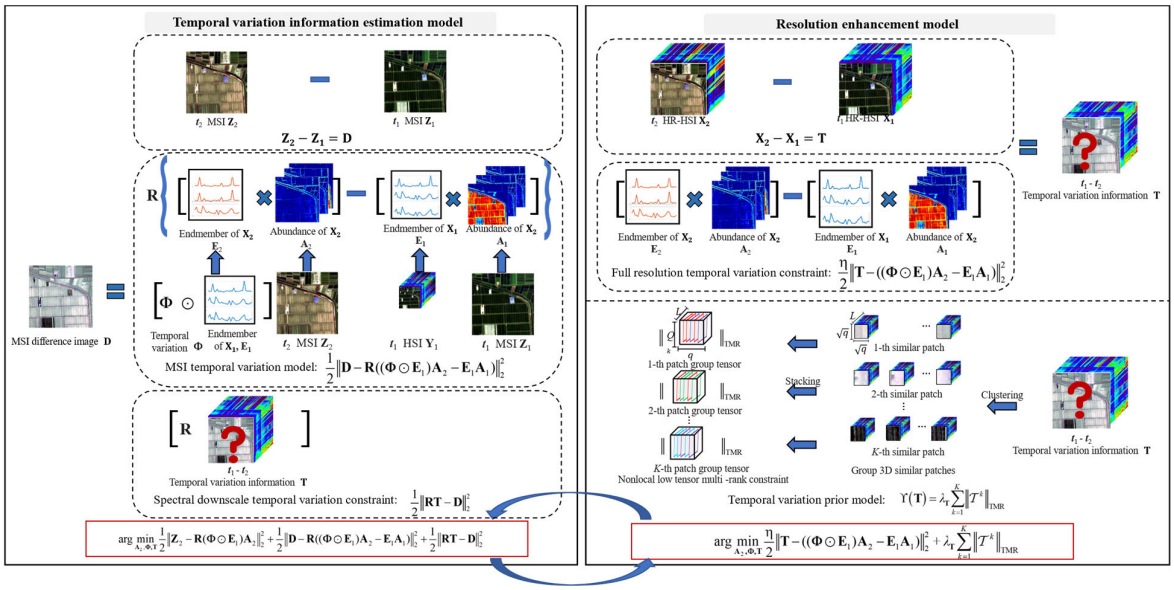


Fig. 1. Framework of the proposed integrated spatial–spectral–temporal fusion method.

fusion images because higher order structural constraints on the temporal variation information are not considered. Fortunately, the temporal variation information between HR-HSI pairs preserves several intrinsic characteristics of HR-HSI. For example, HR-HSI often demonstrates nonlocal spatial similarity and exhibits spectral redundancy and continuity. The temporal variation information of HR-HSI pairs exhibits very similar variations. Similar land cover types tend to undergo comparable temporal variations within a specific time interval in the same natural scene. Since neighboring spectral bands of HR-HSI demonstrate high similarity, the temporal variations reflected in these similar bands are also highly correlated. Therefore, we can incorporate this prior knowledge into the regularization term of the temporal variation information to enhance the accuracy of recovering the missing information. Consequently, the recovered information from the temporal variation information improves the reliability of estimating the information of the target image.

With the assumption, the spatial blur matrix $\mathbf{F}$, the spatial downsampling matrix $\mathbf{S}$, and the spectral response function $\mathbf{R}$ have been accurately estimated from the observed data or known from previous work [17]. The fusion problem is then converted to estimate the temporal variability $\mathbf{\Phi}$, the abundance matrix $\mathbf{A}_2$, and the time variation image $\mathbf{T}$. We formulate the fusion problem as an optimization problem

$$\begin{aligned}\arg\min_{\mathbf{A}_2,\mathbf{\Phi},\mathbf{T}} &\frac{1}{2}\|\mathbf{Z}_2-\mathbf{R}(\mathbf{\Phi}\odot\mathbf{E}_1)\mathbf{A}_2\|_2^2+\frac{1}{2}\|\mathbf{RT}-\mathbf{D}\|_2^2\\&+\frac{1}{2}\|\mathbf{D}-\mathbf{R}((\mathbf{\Phi}\odot\mathbf{E}_1)\mathbf{A}_2-\mathbf{E}_1\mathbf{A}_1)\|_2^2\\&+\frac{\eta}{2}\|\mathbf{T}-((\mathbf{\Phi}\odot\mathbf{E}_1)\mathbf{A}_2-\mathbf{E}_1\mathbf{A}_1)\|_2^2\\&+\Psi(\mathbf{A}_2)+\Xi(\mathbf{\Phi})+\Upsilon(\mathbf{T})\end{aligned}\tag{13}$$

where parameter η is the contribution of the temporal variation information term to the objective function, and in this article, we empirically set to 0.3. Since problem (13) is an ill-posed problem, it is necessary to introduce appropriate prior information on the variables as regularization terms to solve this problem. For this, we define three regularization term functions $\mathbf{\Psi}(\mathbf{A}_2)$, $\mathbf{\Xi}(\mathbf{\Phi})$, and $\Upsilon(\mathbf{T})$ corresponding to variables $\mathbf{A}_2$, $\mathbf{\Phi}$, and $\mathbf{T}$, respectively. We give the details of the regularization function in the following.

In HR-HSI, natural scenes typically consist of only a few material types. The abundance fractions associated with these materials often have a sparse overall distribution, with many nonzero elements. Moreover, neighboring pixels are likely to exhibit similar material characteristics and correspond to the same material endmember, indicating that they should have the same abundance fraction. Therefore, we promote the sparsity of abundance fractions in horizontal and vertical gradients by using a mixed $L_{2,1}$ norm, achieving sparsity at global and individual levels

$$\Psi(\mathbf{A}_2)=\frac{\lambda_{\mathbf{A}}}{2}\left(\|H_h\mathbf{A}_2\|_{2,1}+\|H_v\mathbf{A}_2\|_{2,1}\right).\tag{14}$$

HR-HSI exhibits high correlation and spectral continuity, implying that spectral variations occurring within HR-HSI should possess similar characteristics. Therefore, we promote this by enforcing that smoothness in unitary matrix constraint helps eliminate trivial solutions

$$\Xi(\mathbf{\Phi})=\frac{\lambda_1}{2}\left\|\mathbf{\Phi}-\mathbf{1}_L\mathbf{1}_{\mathbf{Q}}\right\|_F^2+\frac{\lambda_2}{2}\|H_L\mathbf{\Phi}\|_F^2.\tag{15}$$

To ensure the accuracy of the generated temporal variation information, we obtained two prior knowledge assumptions for the boundaries of the temporal variation information $\mathbf{T}$. Similar to the nonlocal similarity that exists in high-resolution HSI of real scenes, it is reasonable to assume that temporal variation information exhibits nonlocal similarity. This is because similar features in the same scene are more likely to undergo the same temporal variations. In addition, HSI exhibits significant redundancy in the spectral dimension. It is presumed that neighboring spectral bands share similar temporal variation

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:25:27 UTC from IEEE Xplore. Restrictions apply.

characteristics, causing the temporal variation information to be highly redundant. Given the spatial and spectral similarity and redundancy of the features, it indicates that the temporal variation information has a low rank. Consequently, incorporating nonlocal similarity and low rankness improves temporal variation estimation accuracy. In previous work, low tensor multirank has shown better performance for HSI super-resolution [38] and denoising [74]. Therefore, we use a nonlocal low tensor multirank prior as a regularization term. The MSI difference image $\mathbf{D}$ preserves the most spatial information of the temporal variation information, so we first learn nonlocal similarity from the MSI difference image $\mathbf{D}$. We use a 3-D tensor, $\mathcal{D} \in \mathbb{R}^{W\times H\times l}$, with $W \times H = N$, to represent the MSI difference image $\mathbf{D}$ folded into a 3-D image. Tensor $\mathcal{D}$ is then partitioned into full-band patches (FBPs) of size $\sqrt{q} \times \sqrt{q} \times l$ and finally use the K-means++ clustering algorithm to cluster these FBPs into K groups, denoted as $\mathbf{D}^k = \{\mathcal{D}^{(k,j)}\}_{j=1}^{N_k}$, with $k = 1, 2, \ldots, K$, where N_k denotes the number of FBPs in the kth group and $\mathcal{D}^{(k,j)}$ represents the jth FBP in the kth group of FBPs. Since the MSI difference image $\mathbf{D}$ can be considered as a spectral downsampled version of the temporal variation information $\mathbf{T}$, the temporal variation information$\mathbf{T}$ should have the same spatial information as $\mathbf{D}$ and retain the same spatial self-similarity. Therefore, by learning the clustering structure of the MSI difference image $\mathbf{D}$, we obtained the grouped FBPs of $\mathbf{T}$, denoted as $T^k = \{\mathcal{T}^{(k,j)} \in \mathbb{R}^{\sqrt{q}\times\sqrt{q}\times L}\}_{j=1}^{N_k}$, where $\mathcal{T}^{(k,j)}$ represents the jth FBP in the kth group of FBPs in the temporal variation information. We stack all the FBPs in the kth group along the spatial dimension and form a 3-D tensor $\mathcal{T}^{(k,j)} \in \mathbb{R}^{N_k\times L\times q}$, which is composed of similar patches clustered in a group and has strong low tensor rank properties. Therefore, we applied the low tensor multirank prior to the group-FBPs of the temporal variation information

$$\Upsilon(\mathbf{T}) = \lambda_{\mathbf{T}} \sum_{k=1}^{K} \left\|\mathcal{T}^k\right\|_{\mathrm{TMR}} \tag{16}$$

where the symbol $\|\cdot\|_{\mathrm{TMR}}$ denotes the tensor multirank [74].

Definition 1 (Tensor Multirank): For a third-order tensor $\mathcal{A} \in \mathbb{R}^{I_1\times I_2\times I_3}$, its horizontal, lateral, and frontal slices are denoted by $\mathcal{A}(i, :, :)$, $\mathcal{A}(:, i, :)$, and $\mathcal{A}(:, :, i)$, respectively. In this article, we denote the kth frontal slice of the third-order tensor $\mathcal{A}(:, :, k)$, more compactly as $\mathcal{A}^{(k)}$. For tensor $\mathcal{A} \in \mathbb{R}^{I_1\times I_2\times I_3}$, $\overline{\mathcal{A}} \in \mathrm{C}^{n_1\times n_2\times n_3}$ denotes the $\mathcal{A}$ result of fast Fourier transform (FFT) on along the 3rd dimension. By using $\mathcal{F}$ denoting FFT command in MATLAB, we can easily get $\overline{\mathcal{A}} = \mathcal{F}(\mathcal{A}, [], 3)$. Conversely, we can use the inverse FFT command $\mathcal{F}^{-1}$ to compute $\mathcal{A}$ from $\overline{\mathcal{A}}$, i.e., $\mathcal{A} = \mathcal{F}^{-1}(\overline{\mathcal{A}}, [], 3)$. The multirank of a tensor $\mathcal{A} \in \mathbb{R}^{I_1\times I_2\times I_3}$ is a vector $\mathbf{p}$ with its 1th element equal to the rank of the first frontal slice of $\overline{\mathcal{A}}$ [74].

Definition 2 (Tensor Nuclear Norm): For a tensor $\mathcal{A} \in \mathbb{R}^{I_1\times I_2\times I_3}$, the tensor kernel norm $\|\mathcal{A}\|_{\mathrm{TNN}}$ is denoted as the sum of the singular values of all frontal slices of $\mathcal{A}$. Let $\mathcal{A} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T$ be the t-SVD of tensor $\mathcal{A}$. The tensor nuclear norm of $\mathcal{A}$ is defined as

$$\|\mathcal{A}\|_{\mathrm{TNN}} = \sum_{i=1}^{I_3}\sum_{j}\sigma_j\left(\overline{\mathbf{A}}^{(i)}\right) = \sum_{i=1}^{I_3}\sum_{i}\overline{\mathcal{S}}(i, i, j) \tag{17}$$

where $\sigma_j(\overline{\mathbf{A}}^{(i)})$ denotes the jth singular value of $\overline{\mathbf{A}}^{(i)}$.

Compared to calculating the sum of singular values in the tensor nuclear norm, low tensor multirank replaces the sum of singular values as a logarithmic sum of singular values and, in some works, low tensor multirank is defined as [75]

$$\|\mathcal{A}\|_{\mathrm{TMR}} = \frac{1}{I_3}\sum_{i=1}^{I_3}\mathrm{LS}\left(\overline{\mathbf{A}}^{(i)}\right) \tag{18}$$

where $\mathrm{LS}(\mathbf{A}) = \sum_i \log(\sigma_i(\mathbf{A}) + \varepsilon)$ is the log sum (LS) of singular values and ε is a very small positive number to ensure that it is not equal to 0. We impose a low tensor multirank constraint on the group-FBPs of temporal variation information, in order to enhance the recovery of temporal variation information.

For the multivariate solution problem defined in (13), the simultaneous solutions are nonconvex for the variables $\mathbf{A}_2$, $\mathbf{\Phi}$, and $\mathbf{T}$ but convex when fixing the other variables to solve one of the variables individually. This motivates us to use the ALS method to minimize the loss function corresponding to each variable in order to obtain a local stationary point. We give the specific steps in Algorithm 1.

A. Optimizing With Respect to A_2

Fixing the variables $\mathbf{\Phi}$ and $\mathbf{T}$ in (13) and selecting the terms in (13) with respect to $\mathbf{A}_2$, we rewrite the optimization problem

$$\begin{aligned}\mathbf{A}_2 = \arg\min_{\mathbf{A}_2} &\frac{1}{2}\|\mathbf{Z}_2 - \mathbf{R}(\mathbf{\Phi}\odot\mathbf{E}_1)\mathbf{A}_2\|_F^2 \\ &+ \frac{1}{2}\|\mathbf{D} - \mathbf{R}((\mathbf{\Phi}\odot\mathbf{E}_1)\mathbf{A}_2 - \mathbf{E}_1\mathbf{A}_1)\|_F^2 \\ &+ \frac{\eta}{2}\|\mathbf{T} - ((\mathbf{\Phi}\odot\mathbf{E}_1)\mathbf{A}_2 - \mathbf{E}_1\mathbf{A}_1)\|_F^2 \\ &+ \frac{\lambda_{\mathbf{A}}}{2}\left(\|H_h\mathbf{A}_2\|_{2,1} + \|H_v\mathbf{A}_2\|_{2,1}\right).\end{aligned} \tag{19}$$

We utilize a partitioning strategy and use the ADMM to solve this variable optimization problem. By defining the auxiliary variables $\mathbf{B}_1 = \mathbf{A}_2$, $\mathbf{B}_2 = H_h(\mathbf{A}_2)$, and $\mathbf{B}_3 = H_v(\mathbf{A}_2)$, (19) can be expressed as

$$\begin{aligned}\mathbf{A}_2 = \arg\min_{\mathbf{A}_2} &\frac{1}{2}\|\mathbf{Z}_2 - \mathbf{R}(\mathbf{\Phi}\odot\mathbf{E}_1)\mathbf{A}_2\|_F^2 \\ &+ \frac{1}{2}\|\mathbf{D} - \mathbf{R}((\mathbf{\Phi}\odot\mathbf{E}_1)\mathbf{A}_2 - \mathbf{E}_1\mathbf{A}_1)\|_F^2 \\ &+ \frac{\eta}{2}\|\mathbf{T} - ((\mathbf{\Phi}\odot\mathbf{E}_1)\mathbf{A}_2 - \mathbf{E}_1\mathbf{A}_1)\|_F^2 \\ &+ \frac{\lambda_{\mathbf{A}}}{2}\left(\|\mathbf{B}_2\|_{2,1} + \|\mathbf{B}_3\|_{2,1}\right).\end{aligned} \tag{20}$$

Let the symbol $\mathrm{vec}(\cdot)$ be the vectorization operation and $\mathbf{u} = \mathrm{vec}(\mathbf{A}_2)$, and we represent the scaled augmented Lagrangian function of (20) as

$$\begin{aligned}&\mathcal{L}(\mathbf{u}, \mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3, \mathbf{V}_1, \mathbf{V}_2, \mathbf{V}_3) \\ &= \frac{1}{2}\left\|\mathrm{vec}(\mathbf{Z}_2) - \mathrm{rep}_M(\mathbf{R}(\Phi\odot\mathbf{E}_1)\mathbf{u})\right\|^2 \\ &\quad + \frac{1}{2}\left\|\mathrm{vec}(\mathbf{D}) - \mathrm{rep}_M(\mathbf{R}((\Phi\odot\mathbf{E}_1)\mathbf{u})) + \mathrm{vec}(\mathbf{E}_1\mathbf{A}_1)\right\|^2\end{aligned}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:25:27 UTC from IEEE Xplore. Restrictions apply.

$$+\frac{\eta}{2}\left\|\text{vec}(\mathbf{T})-\text{rep}_M((\Phi\odot\mathbf{E}_1)\mathbf{u})+\text{vec}(\mathbf{E}_1\mathbf{A}_1)\right\|^2$$
$$+\lambda_A\left(\|\mathbf{B}_2\|^2+\|\mathbf{B}_3\|^2\right)$$
$$+\frac{\rho}{2}\left(\|\text{vec}(\mathbf{B}_1)-\mathbf{u}+\mathbf{V}_1\|^2\right.$$
$$+\|\text{vec}(\mathbf{B}_2)-H_h\text{vec}(\mathbf{B}_1)+\mathbf{V}_2\|^2$$
$$\left.+\|\text{vec}(\mathbf{B}_3)-H_v\text{vec}(\mathbf{B}_1)+\mathbf{V}_3\|^2\right) \quad (21)$$

where $\mathbf{V}_i$, $i=1,2,3$ are the dual variable, ρ is the step size, with $\rho>0$, and $\text{rep}_M(V)$ is the block diagonal matrix for matrix V that is represented as

$$\text{rep}_M(\mathbf{V})=\begin{bmatrix}\mathbf{V}&0&\cdots&0\\0&\mathbf{V}&&0\\\vdots&&\ddots&\vdots\\0&0&\cdots&\mathbf{V}\end{bmatrix}. \quad (22)$$

The details of the optimization of the respective variables in (21) will be given in Appendix A.

B. Optimizing With Respect to $\boldsymbol{\Phi}$

With fixed variables $\mathbf{A}_2$ and $\mathbf{T}$, and selecting the term in (13) associated with $\boldsymbol{\Phi}$, the optimization problem for variable $\boldsymbol{\Phi}$ can be formulated as

$$\boldsymbol{\Phi}=\arg\min_{\boldsymbol{\Phi}}\frac{1}{2}\|\mathbf{Z}_2-\mathbf{R}(\boldsymbol{\Phi}\odot\mathbf{E}_1)\mathbf{A}_2\|_F^2$$
$$+\frac{1}{2}\|\mathbf{D}-\mathbf{R}((\boldsymbol{\Phi}\odot\mathbf{E}_1)\mathbf{A}_2-\mathbf{E}_1\mathbf{A}_1)\|^2$$
$$+\frac{\eta}{2}\|\mathbf{T}-((\boldsymbol{\Phi}\odot\mathbf{E}_1)\mathbf{A}_2-\mathbf{E}_1\mathbf{A}_1)\|_F^2$$
$$+\frac{\lambda_1}{2}\left\|\boldsymbol{\Phi}-\mathbf{1}_L\mathbf{1}_\mathbf{Q}\right\|_F^2+\frac{\lambda_2}{2}\|H_L\boldsymbol{\Phi}\|_F^2. \quad (23)$$

By defining the auxiliary variable $B=\boldsymbol{\Phi}\odot\mathbf{E}_1$, rewrite (23) as

$$\boldsymbol{\Phi}=\arg\min_{\boldsymbol{\Phi}}\frac{1}{2}\|\mathbf{Z}_2-\mathbf{RBA}_2\|_F^2$$
$$+\frac{1}{2}\|\mathbf{D}-\mathbf{R}(\mathbf{BA}_2-\mathbf{E}_1\mathbf{A}_1)\|_F^2$$
$$+\frac{\eta}{2}\|\mathbf{T}-(\mathbf{BA}_2-\mathbf{E}_1\mathbf{A}_1)\|_F^2$$
$$+\frac{\lambda_1}{2}\left\|\boldsymbol{\Phi}-\mathbf{1}_L\mathbf{1}_Q\right\|_F^2+\frac{\lambda_2}{2}\|H_L\boldsymbol{\Phi}\|_F^2. \quad (24)$$

Then, by introducing the dual variable $\mathbf{V}$, the scaled augmented Lagrangian function of (24) can be written as

$$\mathcal{L}(B,\boldsymbol{\Phi},V)=\frac{1}{2}\|\mathbf{Z}_2-\mathbf{RBA}_2\|_F^2$$
$$+\frac{1}{2}\|\mathbf{D}-\mathbf{R}(\mathbf{BA}_2-\mathbf{E}_1\mathbf{A}_1)\|_F^2$$
$$+\frac{\eta}{2}\|\mathbf{T}-(\mathbf{BA}_2-\mathbf{E}_1\mathbf{A}_1)\|_F^2$$
$$+\frac{\lambda_1}{2}\left\|\boldsymbol{\Phi}-\mathbf{1}_L\mathbf{1}_Q\right\|_F^2+\frac{\lambda_2}{2}\|H_L\boldsymbol{\Phi}\|_F^2$$
$$+\frac{\boldsymbol{\rho}}{2}\left(\|B-\boldsymbol{\Phi}\odot\mathbf{E}_1+\mathbf{V}\|_F^2\right). \quad (25)$$

The details of the optimization of each variable in problem (25) are described in Appendix B.

C. Optimizing With Respect to $\boldsymbol{T}$

By fixing the variables $\mathbf{A}_2$ and $\boldsymbol{\Phi}$, and selecting terms from (13) with respect to $\mathbf{T}$, the optimization problem for $\mathbf{T}$ can be written as

$$\mathbf{T}=\arg\min_{\mathbf{T}}\frac{\eta}{2}\|\mathbf{T}-((\boldsymbol{\Phi}\odot\mathbf{E}_1)\mathbf{A}_2-\mathbf{E}_1\mathbf{A}_1)\|_F^2$$
$$+\frac{1}{2}\|\mathbf{RT}-\mathbf{D}\|_F^2+\lambda_\mathbf{T}\sum_{k=1}^{K}\left\|\mathcal{T}^k\right\|_{\text{TMR}}. \quad (26)$$

We can reformulate the optimization problem (26) by introducing the auxiliary variable $\mathcal{B}=\mathcal{T}$ as

$$\mathbf{T}=\arg\min_{\mathbf{T}}\frac{\eta}{2}\|\mathbf{T}-((\boldsymbol{\Phi}\odot\mathbf{E}_1)\mathbf{A}_2-\mathbf{E}_1\mathbf{A}_1)\|_F^2$$
$$+\frac{1}{2}\|\mathbf{RT}-\mathbf{D}\|_F^2+\lambda_\mathbf{T}\sum_{k=1}^{K}\left\|\mathcal{B}^k\right\|_{\text{TMR}}. \quad (27)$$

Similarly, we give the scaled augmented Lagrangian function of (27) as

$$\mathcal{L}(\mathcal{T},\mathcal{B},\mathcal{V})=\frac{1}{2}\|\mathbf{RT}-\mathbf{D}\|_F^2$$
$$+\frac{\eta}{2}\|\mathbf{T}-((\Phi\odot\mathbf{E}_1)\mathbf{A}_2-\mathbf{E}_1\mathbf{A}_1)\|_F^2$$
$$+\frac{\boldsymbol{\rho}}{2}\left(\|\mathcal{B}-\mathcal{T}+\mathcal{V}\|_F^2\right)+\lambda_\mathbf{T}\sum_{k=1}^{K}\left\|\mathcal{B}^k\right\|_{\text{TMR}} \quad (28)$$

where $\mathcal{V}$ is a dual variable. The optimization of $\mathcal{L}(\mathcal{B},\mathcal{T},\mathcal{V})$ with respect to variables $\mathcal{B}$, $\mathcal{T}$, and $\mathcal{V}$ in problem (28) are described in Appendix C.

IV. Experiment

A. Experiments Dataset

The first real dataset is the Yellow River Estuary dataset shown in Fig. 2(a), where the low spatial and coarse temporal resolution HS data and the high spatial and fine temporal resolution MS data are from the ZY-1 02D HS sensor and the Sentinel-2 MS sensor, respectively. Due to the water vapor band and bad band images, 144 of the 166 ZY-1 02D original bands with a spatial resolution of 30 m were selected as the input HSI with a spectral coverage of 400–2500 nm and a spatial size of 150 × 150. Since the Sentinel-2 MSI bands have different spatial resolutions, the second, third, fourth, and eighth bands with 10-m spatial resolution were used as input MSI containing the same region of the HSI in the range of 450 × 450. The fusion task is described in Table I. The goal is to predict images with high spatial, spectral, and fine temporal resolution, on February 10, 2020, based on two Sentinel-2 MSI observed on December 8, 2019, and February 10, 2020, and ZY-1 02D HSI observed on December 11, 2019 (due to satellite revisit cycle limitations, the ZY-1 02D HSI on December 8, 2019, is missing). To evaluate the fusion results, we used the ZY-1 02D HSI observed on February 12, 2020, as the spectral reference image for the fusion results, and the Sentinel-2 MSI observed on February 10, 2020, was used as the spatial reference image for the fusion results. Significant phenological changes and changes in land cover types occurred in the region of interest within the two MSI time intervals on the Yellow River Estuary dataset.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:25:27 UTC from IEEE Xplore. Restrictions apply.

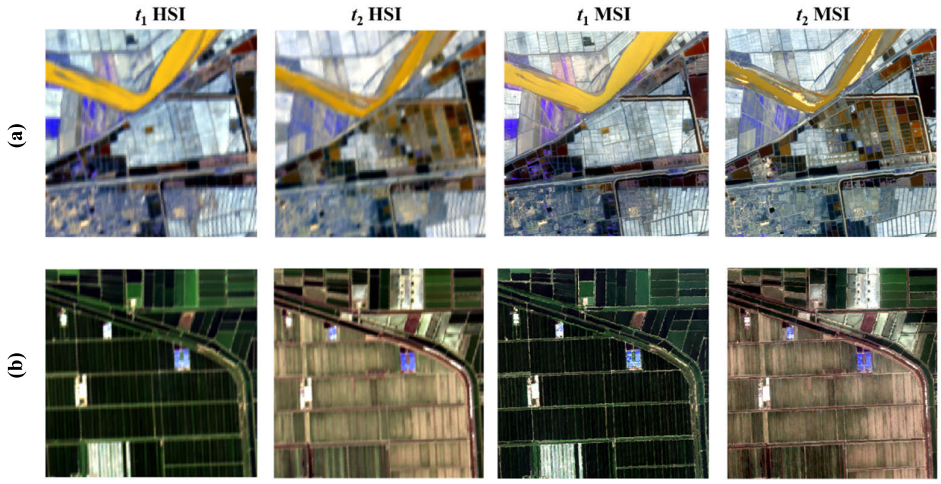


Fig. 2. Experimental datasets. (a) Yellow River Estuary dataset. (b) Yancheng dataset.

TABLE I
CHARACTERISTIC OF EXPERIMENTS DATASETS

Dataset	Sensor	T1	T2	Bands number	Spatial resolution (m)	Spatial Dimensions
Yellow River estuary	ZY-1 02D	December 11, 2019	—	144	30	150 x 150
	Sentinel-2	December 8, 2019	February 10, 2020	4	10	450 x 450
	Fused image	—	February 10, 2020	144	10	450 x 450
Yancheng	ZY-1 02D	September 5, 2020	—	141	30	200 x 200
	Sentinel-2	September 6, 2020	February 22, 2021	4	10	600 x 600
	Fused image	—	February 22, 2021	141	10	600 x 600

The second real dataset is the Yancheng dataset shown in Fig. 2(b), consisting of HSI and MSI data from the ZY-1 02D HS sensor and the Sentinel-2 MS sensor. We selected the ZY-1 02D HSI within a region of 200 × 200 and retained 141 out of the original 166 bands as the input HSI. The input MSI covering the same region had a data dimension of 600 × 600 × 4. The fusion tasks are presented in Table I. We used the ZY-1 02D HSI observed on September 5, 2020, and the Sentinel-2 MS image observed on September 6, 2020, as the input data at observation time $\boldsymbol{t}_1$. The Sentinel-2 MSI was observed on February 22, 2021, as the input data at target time $\boldsymbol{t}_2$, to predict high spatial, spectral, and fine temporal resolution target image data on February 22, 2021. We used the ZY1-02D HSI observed on February 21, 2021, which was the closest date to the predicted date, as the spectral reference image for evaluating the target fusion image, and the Sentinel-2 MSI at $\boldsymbol{t}_2$ as the spatial reference image. The Yancheng dataset showed significant phenological and minor changes in land cover types during the two MSI observation intervals.

B. Compared Methods and Metrics

To validate the effectiveness of our proposed method, five state-of-the-art methods were selected for comparison, including the unsupervised STSF method U-STSF [10], the GLMM-STSF [71], the spectral enhancement method (SREM) [76], the spectral sparse reconstruction method A+ [77], and the spectral super-resolution method J-Slol [78].

However, it is important to note the lack of real data with high spatial–spectral resolution at the desired time for use as ground truth in evaluating the fusion results. To address this limitation, we use the scenario of simultaneous existence of MSI and HSI as spatial and spectral reference images of the fusion results since it is reasonable to assume that the spectral information of the target image is stored in the HSI at the prediction time, while the spatial information of the target is stored in the MSI at the prediction time.

For comprehensively evaluating the fusion results, the spectral angle mapper (SAM), the relative mean square error (RMSE), the relative dimensionless global error in synthesis (ERGAS), the peak-signal-to-noise ratio (PSNR), the correlation coefficient (CC), and the structural similarity (SSIM) were selected as quality evaluation indicators. Specifically, for spectral fidelity, the fused image results were subjected to spatial blurring and downsampling using (2) to generate a spatially downsampled version of the fused image. The blurring and downsampling matrix operators were generated using a 5 × 5 Gaussian kernel with a standard deviation of 2.0, and the vertical and horizontal downsampling rates were both 3. The ideal time observed HSI was used as the spectral reference image, and the SAM, RMSE, ERGAS, and PSNR indicators were selected to evaluate the spatially downsampled version of the fused image. For spatial fidelity, the fusion results of our proposed method and other comparative methods were used to generate a spectral downsampled version of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:25:27 UTC from IEEE Xplore. Restrictions apply.

TABLE II
QUANTITATIVE RESULTS OF THE FUSION RESULTS FOR THE YELLOW RIVER ESTUARY EXPERIMENTS

Methods	Yellow River estuary						
	CC	SSIM	RMSE	SAM	PSNR	ERGAS	Times(s)
SREM	0.443	0.997	0.241	14.34	6.60	144.16	12915.3
A+	0.950	**0.999**	0.246	16.17	15.54	13.30	**2.1**
J-Slol	0.938	0.998	0.163	14.33	16.95	9.47	1323.1
GLMM-STSF	0.928	0.998	0.220	15.40	15.34	12.46	888.7
U-STSF	0.8181	0.997	0.158	16.73	**17.43**	11.47	2.6
CTVRE-STSF	**0.958**	0.996	**0.156**	**10.40**	17. 20	**9.02**	3066.9

TABLE III
QUANTITATIVE RESULTS OF THE FUSION RESULTS FOR THE YANCHNEG EXPERIMENTS

Methods	Yancheng						
	CC	SSIM	RMSE	SAM	PSNR	ERGAS	Times(s)
SREM	0.474	0.973	0.441	32.06	8.73	20.703	22949.1
A+	0.607	0.992	0.262	21.30	11.81	16.02	8.4
J-Slol	0.761	0.990	0.263	25.59	13.37	19.01	3028.4
GLMM-STSF	0.794	0.993	**0.184**	27.82	11.25	18.11	1347.9
U-STSF	0.747	**0.998**	0.270	20.01	14.22	18.74	**3.2**
CTVRE-STSF	**0.860**	0.976	0.219	**14.68**	**14.27**	**16.02**	6156.2

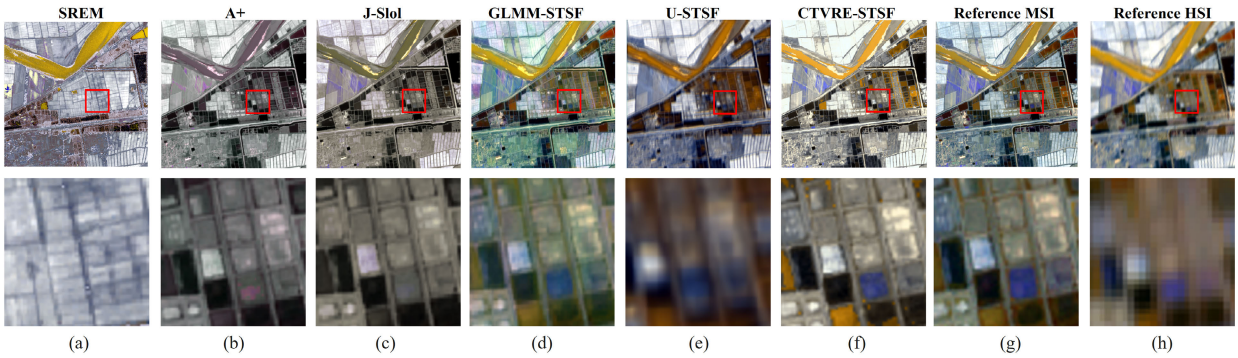


Fig. 3. First row shows the fusion results of the proposed method and competing methods on the Yellow River Estuary dataset. Second row shows the enlarged marked area. (a) Predicted fused image for February 10, 2021 by SREM. (b) Predicted fused image for February 10, 2021 by A+. (c) Predicted fused image for February 10, 2021 by J-Slol. (d) Predicted fused image for February 10, 2021 by GLMM-STSF. (e) Predicted fused image for February 10, 2021 by U-STSF. (f) Predicted fused image for February 10, 2021 by CVTRE-STSF. (g) Sentinel-2 MSI on February 10, 2021. (h) ZY-1 02D HSI on February 12, 2021.

fused image using the spectral response function based on (3), and the target time observed MSI was used as the spatial reference image. The CC and SSIM indicators were calculated to evaluate the spectral downsampled version of the fused image. The computation time (in seconds) was also calculated to evaluate the performance of these methods. We highlight the best results in Tables II and III, which are indicated in bold.

C. Comparison Results on the Yellow River Estuary Dataset

The metrics evaluation of the fusion results of our method and the other four methods on the Yellow River Estuary dataset are shown in Table II. From Table II, we can find that our method achieves the best performance in four metrics, including CC, RMSE, SAM, and ERGAS. In addition, our method achieves significantly better performance in the SAM metric, which indicates that the temporal variation information model we employed can play an important contribution to the spectral fidelity of the target fused image. In Fig. 3, we show the true-color synthetic images of the fusion results of our proposed method and the five compared methods, and in Fig. 3, we can see that the SREM fusion results do not achieve the prediction of the change of the feature at t_2, in which the wrong feature information appears. The other five methods achieve the prediction of feature variation between t_1 and t_2, and visually, the fusion results of the U-STSF method appear blurred; in the enlarged image below, it is clear that it is closer to the low spatial resolution HSI and does not achieve spatial resolution enhancement. In A+, J-Slol, GLMM-STSF, and our CTVRE-STSF all achieve good visual performance, preserving the detailed information of the target fused image well. The enlarged image in the bottom row shows that A+, J-Slol, GLMM-STSF, and our CVTRE-STSF all retain the spatial structure information in the spatial reference image [see Fig. 3(g)] better. In the spectral domain, GLMM-STSF, U-STSF, and our CTVRE-STSF method retain the spectral information better, but GLMM-STSF shows artifacts, and our method is the closest to the reference image spectral. Fig. 4

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:25:27 UTC from IEEE Xplore. Restrictions apply.

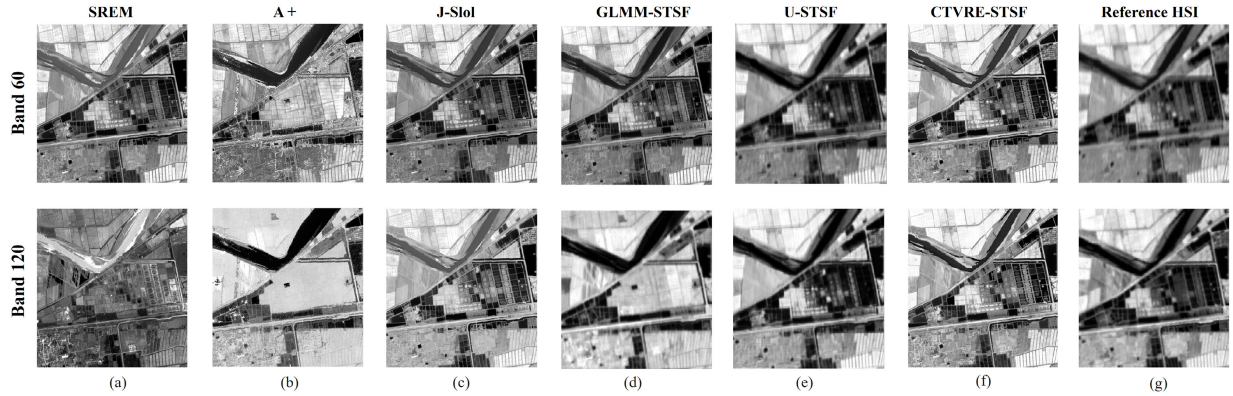


Fig. 4. (Top row) 60th band of the fusion results on the Yellow River Estuary dataset. (bottom row) 120th band of the fusion results.

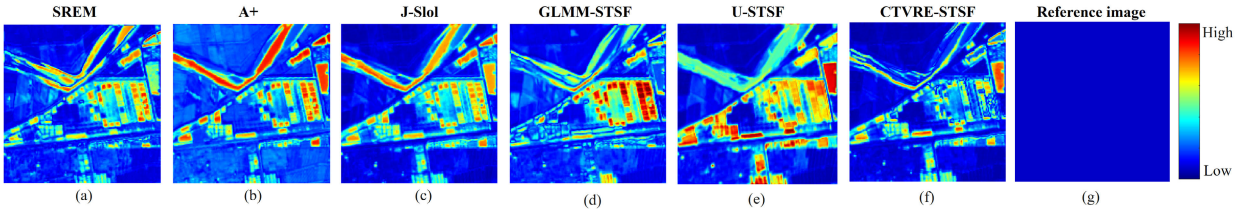


Fig. 5. SAM error images of different fusion results on Yellow River Estuary dataset.

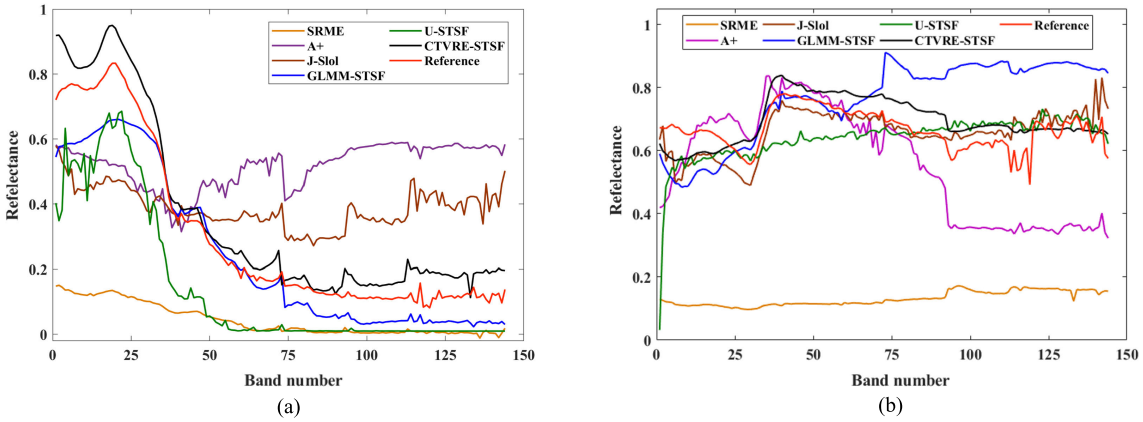


Fig. 6. Spectral curves of the fusion result on (a) pixel (33, 15) and (b) pixel (60, 39) in the Yellow River Estuary dataset.

shows the 60th and 120th bands of fusion results, and it can be seen from Fig. 4 that SREM, J-SIol, GLMM-STSF, U-STSF, and CVTRE-STSF demonstrate better visual effects in the 60th band, and in the 120th band, SREM [Fig. 4(a)] shows incorrect colors, A+ [Fig. 4(b)] failed to correctly predict feature changes, GLMM-STSF [Fig. 4(f)] and U-STSF [Fig. 4(e)] becomes spatially blurred, and only our method [Fig. 4(f)] and J-SIol [Fig. 4(c)] show closer to the reference HSI [Fig. 4(g)]. Fig. 5 shows the SAM error distribution of the fusion results, from which it can be seen that for all methods the spectral distortion is more evident in the river region and in the central region where the land cover type changes. Specifically, compared to the reference image [Fig. 5(g)], the fusion results from SREM [Fig. 5(a)], A+ [Fig. 5(b)], and J-SIol [Fig. 5(c)] show notable spectral distortion in the river region, and the fusion results from GLMM-STSF [Fig. 5(d)] and U-STSF [Fig. 5(e)] demonstrate a higher degree of spectral distortion in the central region of the image, coinciding with areas of substantial land cover variation. Notably, our method achieves improved spectral fidelity [Fig. 5(f)], especially in the river and central regions where the land cover type exhibits significant changes. Fig. 6 shows the comparison of the two typical pixels' spectral curves, and it can be seen from Fig. 6 that the spectral curves of our proposed method (red) are closest to the reference spectral curve (black), which preserves the structure of the spectral curve well.

D. Comparison Results on the Yancheng Dataset

The quantitative evaluation of the fusion results on the Yancheng dataset for the proposed and competing methods is shown in Table III. From Table III, it can be found that our proposed method achieves the best performance for the indicators CC, SAM, PSNR, and ERGAS, especially for SAM, which is significantly lower than the fusion results of the other five methods, indicating that our proposed method can achieve good performance in terms of spectral fidelity. On the indicator SSIM, our proposed method obtains a relatively low performance, which indicates that our proposed method has a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:25:27 UTC from IEEE Xplore. Restrictions apply.

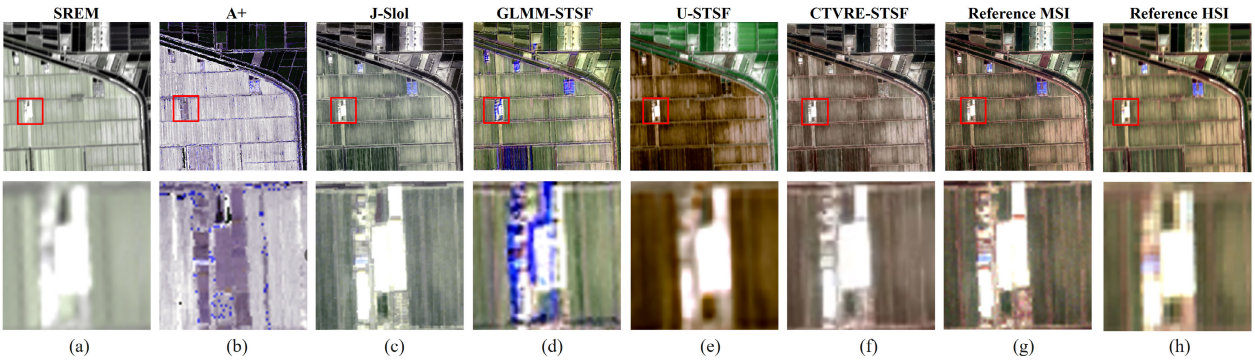


Fig. 7. First row shows the fusion results of the proposed method and competing methods on the Yancheng dataset. Second row shows the enlarged marked area. (a) Predicted fused image for February 22, 2021 by SREM. (b) Predicted fused image for February 22, 2021 by A+. (c) Predicted fused image for February 22, 2021 by J-Slol. (d) Predicted fused image for February 22, 2021 by GLMM-STSF. (e) Predicted fused image for February 22, 2021 by our U-STSF. (f) Predicted fused image for February 22, 2021 by our CVTRE-STSF. (g) Sentinel-2 MSI on February 22, 2021. (h) ZY-1 02D HSI on February 21, 2021.

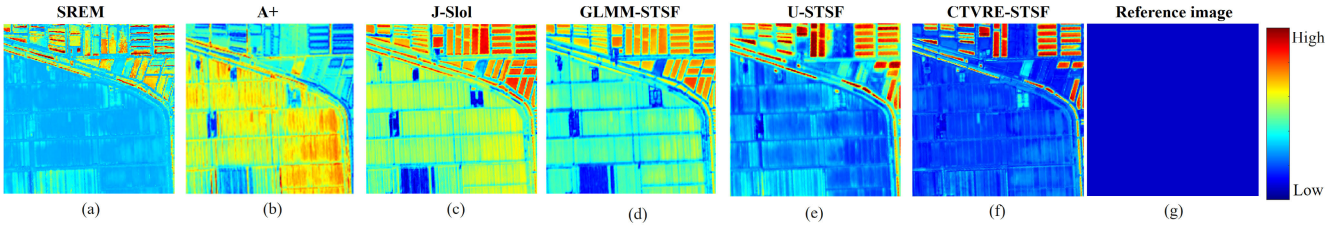


Fig. 8. SAM error images of different fusion results on the Yancheng dataset.

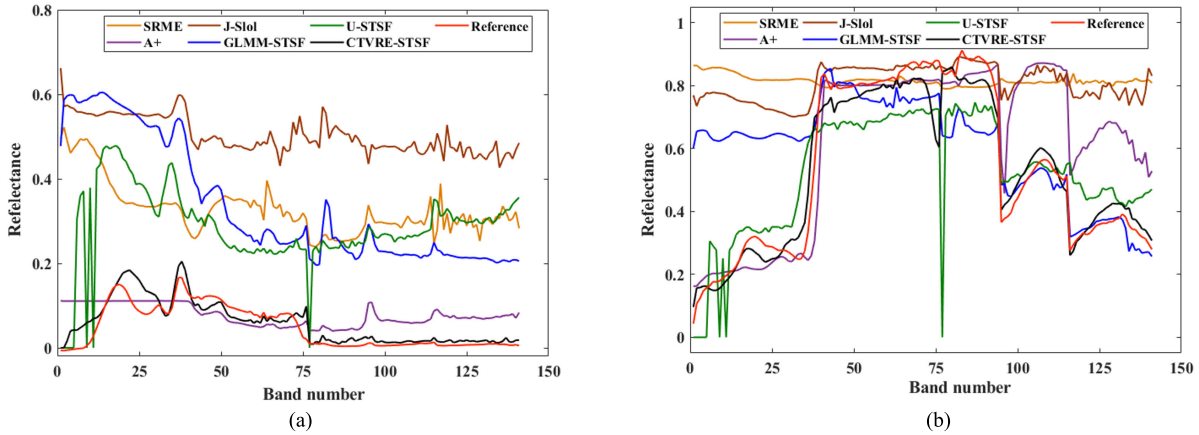


Fig. 9. Spectral curves of the fusion results on (a) pixel (15, 87) and (b) pixel (171, 99) in the Yancheng dataset.

general of the spatial structure, probably because, although the nonlocal similarity prior we introduce in the temporal variation information is able to preserve the spatial and spatial structure information of the temporal variation information, the constructed time variation image still has some uncertainties, and these uncertainties on the fusion results are manifested in the corruption of the spatial information of the fused images. We show the true-color synthetic images in Fig. 7 along with a zoomed-in view of the region of interest. For the spectral domain, in general, all methods exhibit more severe spectral distortions, especially in the upper right region, where all methods exhibit severe spectral distortions Specifically, SREM and A+ lose almost all spectral information, GLMM-STSF and U-STSF have erroneous spectra in the upper right region, and J-Slol and our CTVRE-STSF are generally relatively closer to the reference image spectra. In terms of spatial fidelity, A+ shows severe pretzel error, and SREM and U-STSF show more blur, as seen in the zoomed image below, which does not implement spatial resolution enhancement. J-Slol and our CTVRE-STSF show better preservation of the spatial structure and details in the reference MSI at the target time. Fig. 8 shows the SAM error distribution of the fusion results. Fig. 8 reveals that, aside from the A+ method [Fig. 8(b)], the top-right corner of the image suffers from the most pronounced spectral distortion when compared to the reference image [Fig. 8(g)]. In contrast, the lower region shows improved spectral fidelity with the SREM [Fig. 8(a)], J-Sol [Fig. 8(c)], GLMM-STSF [Fig. 8(d)], U-STSF [Fig. 8(e)], and our CTVRE-STSF [Fig. 8(f)] methods, relative to the upper region. Notably, our CTVRE-STSF method demonstrates the least spectral distortion overall. Fig. 9 shows the spectral curves of two typical pixels in the fusion results,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:25:27 UTC from IEEE Xplore. Restrictions apply.

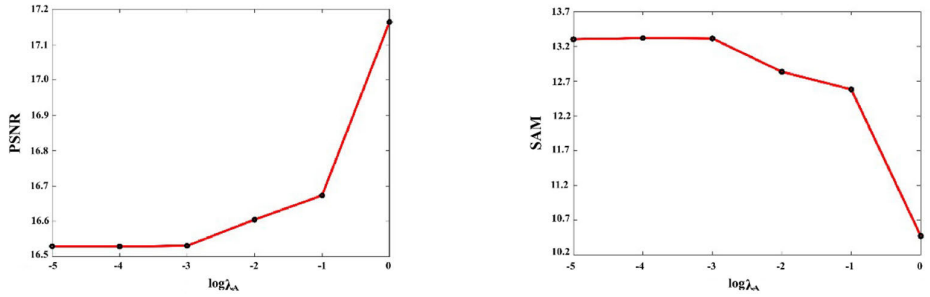


Fig. 10. Effect of the parameter λ_A on the performance of the proposed method.

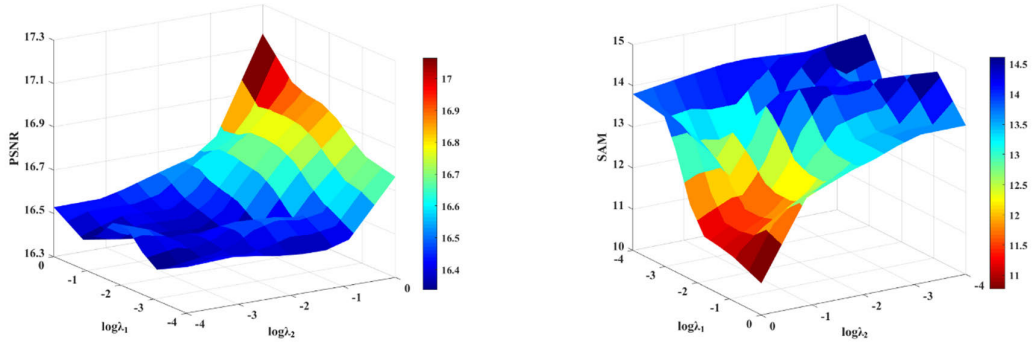


Fig. 11. Effect of the parameter λ_1 and λ_2 on the performance of the proposed method.

and it is clear from Fig. 9(a) and (b) that the CTVRE-STSF method shows excellent performance in terms of spectral preservation.

E. Parameters Discussion

In our method, there are more parameters, including the regularization parameters λ_A in the abundance subproblem, the regularization parameters λ_1 and λ_2 in the temporal variation subproblem, the number of clusters K, the spatial dimension $\sqrt{q}$ of the FBPs, and the regularization parameter λ_T in the temporal variation information subproblem. Inevitably, the impact of different parameters on the experiment should be investigated. To assess the sensitivity of the proposed method to these parameters, we divided these parameters into three groups according to the subproblem in which they are contained. We take the Yellow River Estuary dataset as a representative, and by controlling the variation of each set of parameters and plotting the PSNR and SAM curves of the corresponding fusion results, we can observe the influence of the parameters on our proposed method.

For the regularization parameter $\lambda_{\mathbf{A}}$ in the abundance subproblem, we set $\log\lambda_{\mathbf{A}}$ to vary from -5 to 0. From Fig. 10, it can be found that when the parameter $\log\lambda_{\mathbf{A}}$ varied from -2 to 0, there is a significant improvement in the metrics PSNR and SAM, and the best values of both PSNR and SAM are achieved at 0. Therefore, in the experiments on both datasets, we set $\log\lambda_{\mathbf{A}} = 0$, i.e., $\lambda_{\mathbf{A}} = \mathbf{1}$.

For the regularization parameters λ_1 and λ_2 in the time variations subproblem, we set λ_1 and λ_2 to vary from -4 to 0 and plot the 3-D map of the indicators PSNR and SAM simultaneously with parameters λ_1 and λ_2, respectively, with the colors from blue to red representing the better PSNR and SAM indicator values (larger PSNR values represent better performance, while smaller SAM values correspond to better performance). From Fig. 11, it can be observed that the indicators PSNR and SAM are gradually getting better overall as λ_1 and λ_2 change from -4 to 0, and the best performance is achieved when $\log\lambda_1 = 0$ and $\log\lambda_2 = 0$ for both indicators PSNR and SAM, 17.06 and 10.77, respectively, so in our experiments, we set $\log\lambda_1 = 0$ and $\log\lambda_2 = 0$, that is, $\lambda_1 = 1$ and $\lambda_2 = 1$.

For the parameters K, $\sqrt{\mathbf{q}}$, and λ_T in the time variation information subproblem, we set K to vary from 1 to 201, $\sqrt{\mathbf{q}}$ to vary from 1 to 11, and λ_T to vary from -3 to 1, and plotted the scatter plots of indicators PSNR and SAM with the three parameters simultaneously. The scatter plots of the PSNR and SAM indicators are plotted with the three parameters simultaneously, and we use both color and scatter size changes to represent the PSNR and SAM changes. The bluer the color and the smaller the scatter means that both indicators are worse, and the redder the color and the larger seen in Fig. 12; the parameter changes in the temporal variations information subproblem have a very obvious effect on the proposed method; and the PSNR and SAM metrics produce drastic changes as the parameters change. Therefore, it is necessary to further analyze the effects of each of these three parameters on the experimental results. Specifically, the patch size $\sqrt{q}$ exhibits a small effect on the experiments, PSNR and SAM metrics change less with the transformation of $\sqrt{q}$ but usually better values appear after $\sqrt{q} \geq 5$. The total number of clusters K also shows a small effect on the experiment, with nonlocal similarity not occurring at $K = 0$. Both PSNR and SAM perform poorly, and better values of the metrics usually occur after $K \geq 51$. The most important effect on the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:25:27 UTC from IEEE Xplore. Restrictions apply.

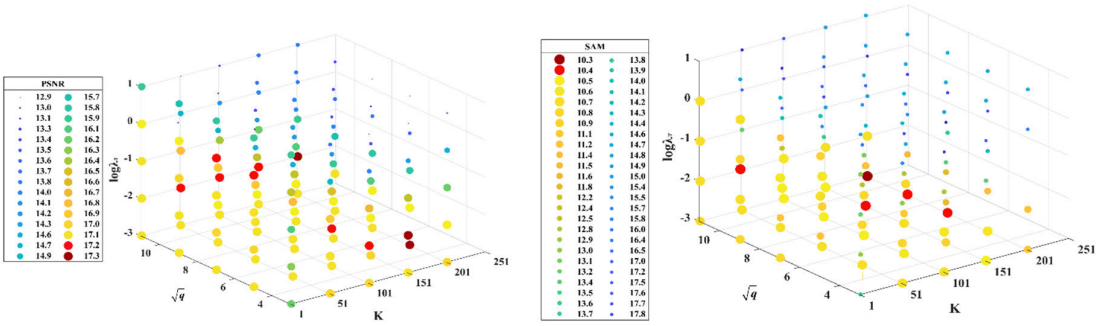


Fig. 12. Effect of the parameter K, $\sqrt{q}$, and λ_T on the performance of the proposed method.

performance of the experiment is shown by the low rankness of the time variations image, and as $\log\lambda_T$ changes from -3 to 1, the PSNR metric becomes better first but declines rapidly after $\log\lambda_T > -2$. The SAM metric becomes relatively worse as $\log\lambda_T$ becomes larger and also declines sharply after $\log\lambda_T > -2$. The best metrics for PSNR and SAM, 17.3 and 10.3, do not appear simultaneously, and the PSNR and SAM achieved the second-best metrics of 17.2 and 10.4 at $K = 51$, $\sqrt{q}$ =11, and $\log\lambda_T = -2$. Finally, we set $K = 51$, $\sqrt{q} =$ 11, and $\log\lambda_T = 0.01$.

F. Execution Times

We give a comparison of the running times of all methods on the two datasets at the end of Tables II and III. Our methods have longer run times since our methods have three subproblems to optimize, in particular, the solution of the nonlocal low tensor multirank constraint in the temporal variation information subproblem.

V. Conclusion

This article presents an STSF method aimed at integrating low spatial and coarse temporal resolution HSI with high spatial and fine temporal resolution MSI to yield the highest spatial, spectral, and fine temporal resolution images. The primary challenge stems from the relatively coarser temporal resolution of HSI, which introduces difficulties in establishing robust constraints to recover high-dimensional spectral information within the target HR-HSI. To tackle this issue, we extend our prior work by developing a spectral downscaling temporal variation information estimation model grounded in the GLMM and MSI differential image. This model effectively estimates temporal variations. In addition, we introduce a resolution enhancement model that employs prior temporal variation information as a regularization term. The combination of these two models gives rise to the CVTRE-STSF approach. The temporal variation information estimation model provides temporal variation estimations, while the resolution enhancement model supplements constraints to enhance the resolution of temporal variation information. In future work, we will explore two aspects to extend our method: one is to capture and model nonlinear temporal variation in the fusion problem and the other is to further investigate the inherent characteristics of the temporal variation information model for more accurate recovery of temporal variation information between HR-HSI.

Appendix A
Optimization of the ADMM for the $\mathbf{A}_2$ Subproblem

In this section, we describe the optimization of each variable in the scaled augmented Lagrangian function (21). The object function in (21) is not convex for all variables simultaneously, but it is a convex problem with respect to individual variables when other variables are fixed. Therefore, we solve (21) by decomposing into multiple manageable subproblems.

A. Optimizing With Respect to $\boldsymbol{u}$

By selecting the term of the minimization function (21) with respect to $\mathbf{u}$, the subproblem of $\mathbf{u}$ can be acquired

$$\begin{aligned}\mathbf{u} \in \arg\min_{\mathbf{u}} \mathcal{L}(\mathbf{u}, \mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3, \mathbf{V}_1, \mathbf{V}_2, \mathbf{V}_3)\\ = \arg\min_{\mathbf{u}} \frac{1}{2}\left\|\text{vec}(\mathbf{Z}_2) - \text{rep}_M(\mathbf{R}(\Phi \odot \mathbf{E}_1)\mathbf{u})\right\|^2\\ + \frac{1}{2}\left\|\text{vec}(\mathbf{D}) - \text{rep}_M(\mathbf{R}((\Phi \odot \mathbf{E}_1)\mathbf{u})) + \text{vec}(\mathbf{E}_1\mathbf{A}_1)\right\|^2\\ + \frac{\eta}{2}\left\|\text{vec}(\mathbf{T}) - \text{rep}_M((\Phi \odot \mathbf{E}_1)\mathbf{u}) + \text{vec}(\mathbf{E}_1\mathbf{A}_1)\right\|^2\\ + \frac{\rho}{2}\|\text{vec}(\mathbf{B}_1) - \mathbf{u} + \mathbf{V}_1\|^2. \quad (29)\end{aligned}$$

Since $\mathbf{u}$ is the vector form of the abundance matrix $\mathbf{A}_2$, we rewrite the problem (29) in pixel form for convenience in solving

$$\begin{aligned}\arg\min_{\mathbf{u}} \sum_{n=1}^{M}\Bigg(\frac{1}{2}\left\|\mathbf{z}_{2\ m,n} - \mathbf{R}(\Phi \odot \mathbf{E}_1)\mathbf{a}_{2,n}\right\|^2\\ + \frac{1}{2}\left\|\mathbf{d}_{m,n} - \left(\mathbf{R}\left((\Phi \odot \mathbf{E}_1)\mathbf{a}_{2,n} - \mathbf{E}_1\mathbf{a}_{1,n}\right)\right)\right\|^2\\ + \frac{\eta}{2}\left\|\mathbf{t}_{m,n} - \left((\Phi \odot \mathbf{E}_1)\mathbf{a}_{2,n} - \mathbf{E}_1\mathbf{a}_{1,n}\right)\right\|^2\\ + \frac{\rho}{2}\left\|\mathbf{b}_{1,n} - \mathbf{a}_{2,n} + \mathbf{v}_{1,n}\right\|^2\Bigg). \quad (30)\end{aligned}$$

Taking the gradient of an and let it be zero, we have

$$\begin{aligned}\big(2(\mathbf{R}(\Phi \odot \mathbf{E}_1))^{\mathsf{T}}\mathbf{R}(\Phi \odot \mathbf{E}_1)\\ + \eta(\Phi \odot \mathbf{E}_1)^{\mathsf{T}}(\Phi \odot \mathbf{E}_1) + \rho\mathbf{I}\big)\mathbf{a}_{2,n}\\ = (\mathbf{R}(\Phi \odot \mathbf{E}_1))^{\mathsf{T}}\big(\mathbf{z}_{2m,n} + \mathbf{d}_{m,n} + \mathbf{E}_1\mathbf{a}_{1,n}\big)\\ + \eta(\Phi \odot \mathbf{E}_1)^{\mathsf{T}}\big(\mathbf{t}_{m,n} + \mathbf{E}_1\mathbf{a}_{1,n}\big) + \rho\big(\mathbf{b}_{1,n} + \mathbf{v}_{1,n}\big). \quad (31)\end{aligned}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:25:27 UTC from IEEE Xplore. Restrictions apply.

Then, the solution is given as

$$\mathbf{a}_n^* = \Omega_1^{-1}\Omega_2 \tag{32}$$

with

$$\begin{aligned}\Omega_1 &= \big(2(\mathbf{R}(\mathbf{\Phi}\odot\mathbf{E}_1))^{\mathbf{T}}\mathbf{R}(\mathbf{\Phi}\odot\mathbf{E}_1)\\ &\quad+\eta(\mathbf{\Phi}\odot\mathbf{E}_1)^{\mathbf{T}}(\mathbf{\Phi}\odot\mathbf{E}_1)+\rho\mathbf{I}\big)\\ \Omega_2 &= (\mathbf{R}(\mathbf{\Phi}\odot\mathbf{E}_1))^{\mathbf{T}}\big(\mathbf{z}_{2m,n}+\mathbf{d}_{m,n}+\mathbf{E}_1\mathbf{a}_{1,n}\big)\\ &\quad+\eta(\mathbf{\Phi}\odot\mathbf{E}_1)^{\mathbf{T}}\big(t_{m,n}+\mathbf{E}_1\mathbf{a}_{1,n}\big)+\rho\big(\mathbf{b}_{1,n}+\mathbf{v}_{1,n}\big).\end{aligned} \tag{33}$$

B. Optimizing With Respect to $\boldsymbol{B}_1$

The subproblem of $\mathbf{B}_1$ can be written as

$$\begin{aligned}\mathbf{B}_1 &\in \arg\min_{\mathbf{B}_1}\mathcal{L}(\mathbf{u},\mathbf{B}_1,\mathbf{B}_2,\mathbf{B}_3,\mathbf{V}_1,\mathbf{V}_2,\mathbf{V}_3)\\ &= \arg\min_{\mathbf{B}_1}\frac{\rho}{2}\big(\|\mathrm{vec}(\mathbf{B}_1)-\mathbf{u}+\mathbf{V}_1\|_2^2\\ &\qquad+\|\mathrm{vec}(\mathbf{B}_2)-H_h\mathrm{vec}(\mathbf{B}_1)+\mathbf{V}_2\|_2^2\\ &\qquad+\|\mathrm{vec}(\mathbf{B}_3)-H_v\mathrm{vec}(\mathbf{B}_1)+\mathbf{V}_3\|_2^2\big)\end{aligned} \tag{34}$$

which is solved by

$$\begin{aligned}\mathrm{vec}(\mathbf{B}_1) = \big(\mathbf{I}+H_h^TH_h+H_v^TH_v\big)^{-1}(\mathbf{u}-\mathbf{V}_1\\ +H_h^T\mathrm{vec}(\mathbf{B}_2)+H_v^T\mathbf{V}_2+H_v^T\mathrm{vec}(\mathbf{B}_3)+H_v^T\mathbf{V}_3\big).\end{aligned} \tag{35}$$

In (35), calculating the inverse of the matrix is difficult due to the difference operators a and b. However, as in [79] and [73], it can be assumed that the difference operators are block circulant so that we can compute them very efficiently in the Fourier domain, the solution in (35) can be computed as

$$\begin{aligned}[\mathcal{B}_1]_{:,:,c} = \mathcal{F}^{-1}\Big(\big(\mathcal{F}\big(\mathcal{A}_{2:,:,c}-[\mathcal{V}_1]_{:,:,c}\big)\\ +\mathcal{F}^*(\boldsymbol{h}_h)\odot\mathcal{F}\big([\mathcal{B}_2]_{:,:,c}+[\mathcal{V}_2]_{:,:,qc}\big)\\ +\mathcal{F}^*(\boldsymbol{h}_v)\odot\mathcal{F}\big([\mathcal{B}_3]_{:,:,c}+[\mathcal{V}_3]_{:,:,c}\big)\big)\\ \oslash\big(\mathbf{1}_\mathbf{W}\mathbf{1}_H^T+|\mathcal{F}(\boldsymbol{h}_h)|^2+|\mathcal{F}(\boldsymbol{h}_v)|^2\big)\Big)\end{aligned} \tag{36}$$

for $c = 1,\ldots,Q$, where $\mathcal{F}$ and $\mathcal{F}^{-1}$ denote the Fourier transform and its inverse transform, respectively, $\mathcal{F}^*$ is the complex conjugate of the Fourier transform, $\oslash$ is the elementwise division, and $|\cdot|$ denotes the modulus operator. The tensors $\mathcal{A}_2,\mathcal{B}_1,\mathcal{B}_2,\mathcal{B}_3,\mathcal{V}_1,\mathcal{V}_2$, and $\mathcal{V}_3$ of size $W\times H\times Q$ are the third-order versions with respect to the variables.

C. Optimizing With Respect to $\boldsymbol{B}_2$

The optimization subproblem of $\mathbf{B}_2$ is

$$\begin{aligned}\mathbf{B}_2 &\in \arg\min_{\mathbf{B}_2}\mathcal{L}(\mathbf{u},\mathbf{B}_1,\mathbf{B}_2,\mathbf{B}_3,\mathbf{V}_1,\mathbf{V}_2,\mathbf{V}_3)\\ &= \arg\min_{\mathbf{B}_2}\frac{\lambda_A}{2}\|H_h\mathbf{A}_2\|_{2,1}\\ &\qquad+\|\mathrm{vec}(\mathbf{B}_2)-H_h\mathrm{vec}(\mathbf{B}_1)+\mathbf{V}_2\|_2.\end{aligned} \tag{37}$$

The solution of the problem (37) is equivalent to the proximal operator of the L_2 norm. The solution can be obtained by using block soft thresholds, which can be expressed as

$$\mathbf{B}_2^* = \mathrm{soft}_{\lambda_A/\rho}(H_h\mathrm{vec}(\mathbf{B}_1)+\mathbf{V}_2) \tag{38}$$

where the soft operator is defined as

$$\mathrm{soft}_a(b) = \begin{cases}\max\left(1-\dfrac{a}{\|b\|},0\right)b, & b\neq 0\\ 0, & b=0.\end{cases} \tag{39}$$

D. Optimizing With Respect to $\boldsymbol{B}_3$

For $\mathbf{B}_3$, the optimization subproblem is

$$\begin{aligned}\mathbf{B}_3 &\in \arg\min_{\mathbf{B}_3}\mathcal{L}(\mathbf{u},\mathbf{B}_1,\mathbf{B}_2,\mathbf{B}_3,\mathbf{V}_1,\mathbf{V}_2,\mathbf{V}_3)\\ &= \arg\min_{\mathbf{B}_3}\frac{\lambda_A}{2}\|\mathbf{H_h}\mathbf{A}_2\|_{2,1}\\ &\qquad+\|\mathrm{vec}(\mathbf{B}_3)-\mathbf{H_h}\mathrm{vec}(\mathbf{B}_1)+\mathbf{V}_2\|_2^2.\end{aligned} \tag{40}$$

Similar to the optimization strategy for problem $\mathbf{B}_2$, the solution is obtained from a soft threshold

$$\mathbf{B}_3^* = \mathrm{soft}_{\lambda_A/\rho}(H_v\mathrm{vec}(\mathbf{B}_1)+\mathbf{V}_3). \tag{41}$$

E. Dual Variables Update

The update for dual variables is given as follows:

$$\begin{aligned}\mathbf{V}_1 &\leftarrow \mathbf{V}_1+\mathrm{vec}(\mathbf{B}_1)-\mathbf{u}\\ \mathbf{V}_2 &\leftarrow \mathbf{V}_2+\mathrm{vec}(\mathbf{B}_2)-H_h\mathrm{vec}(\mathbf{B}_1)\\ \mathbf{V}_3 &\leftarrow \mathbf{V}_3+\mathrm{vec}(\mathbf{B}_3)-H_v\mathrm{vec}(\mathbf{B}_1).\end{aligned} \tag{42}$$

APPENDIX B
OPTIMIZATION OF THE ADMM FOR THE $\boldsymbol{\Phi}$ SUBPROBLEM

In this section, we describe the optimization of the scaled augmented Lagrangian function (25) with respect to the variables of the time variation.

A. Optimizing With Respect to $\boldsymbol{B}$

By extracting the term in the function (25) with respect to $\mathbf{\Phi}$, the optimization subproblem of $\mathbf{\Phi}$ can be obtained

$$\begin{aligned}B &\in \arg\min_B\mathcal{L}(B,\mathbf{\Phi},V)\\ &= \arg\min_B\frac{1}{2}\|\mathbf{Z}_2-\mathbf{RBA}_2\|_F^2\\ &\qquad+\frac{1}{2}\|\mathbf{D}-\mathbf{R}(\mathbf{BA}_2-\mathbf{E}_1\mathbf{A}_1)\|_F^2\\ &\qquad+\frac{\eta}{2}\|\mathbf{T}-(\mathbf{BA}_2-\mathbf{E}_1\mathbf{A}_1)\|_F^2\\ &\qquad+\frac{\rho}{2}\big(\|B-\mathbf{\Phi}\odot\mathbf{E}_1+\mathbf{V}\|_F^2\big).\end{aligned} \tag{43}$$

Taking the derivative for B and setting it equal to zero, the optimization problem in (43) can be written as

$$\begin{aligned}\big(2\mathbf{R}^T\mathbf{R}+\eta\big)\mathbf{BA}_2\mathbf{A}_2^T+\rho B\\ =\mathbf{R}^T\mathbf{Z}_2\mathbf{A}_2^T+\mathbf{R}^T\mathbf{DA}_2^T+\mathbf{R}^T\mathbf{RE}_1\mathbf{A}_1\mathbf{A}_2^T\\ +\eta\big(\mathbf{TA}_2^T+\mathbf{E}_1\mathbf{A}_1\mathbf{A}_2^T\big)+\rho(\mathbf{\Phi}\odot\mathbf{E}_1-\mathbf{V}).\end{aligned} \tag{44}$$

Equation (44) forms the Sylvester equation AXB − X + C = 0. The equation can be solved in MATLAB using the dlayp, with X = dlayp (A, B, C).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:25:27 UTC from IEEE Xplore. Restrictions apply.

B. Optimizing With Respect to $\boldsymbol{\Phi}$

The optimizing subproblem with respect to $\boldsymbol{\Phi}$ is written as

$$\begin{aligned}\boldsymbol{\Phi} &\in \arg\min_{\boldsymbol{\Phi}} \mathcal{L}(\mathbf{B}, \boldsymbol{\Phi}, \mathbf{V}) \\ &= \arg\min_{P} \frac{\lambda_1}{2}\left\|\boldsymbol{\Phi} - \mathbf{1}_L\mathbf{1}_Q\right\|_F^2 \\ &\quad + \frac{\lambda_2}{2}\|H_L\boldsymbol{\Phi}\|_F^2 + \frac{\boldsymbol{\rho}}{2}\left(\|\mathbf{B} - \boldsymbol{\Phi}\odot\mathbf{E}_1 + \mathbf{V}\|_F^2\right).\end{aligned} \tag{45}$$

Let $\partial\boldsymbol{\mathcal{L}}/\partial\boldsymbol{\Phi}$ equal to zero, and the optimization problem (45) can be written as

$$\begin{aligned}&\left(\lambda_1 + \lambda_2 H_L^T H_L\right)\Phi + \rho\mathbf{E}_1\odot\mathbf{E}_1\odot\boldsymbol{\Phi} \\ &\qquad = \lambda_1\mathbf{1}_L\mathbf{1}_Q + \rho(\mathbf{E}_1\odot\mathbf{B} + \mathbf{E}_1\odot\mathbf{V}).\end{aligned} \tag{46}$$

By rewriting (46) in the form of a vector, it can be readily solved

$$\begin{aligned}&\left(\lambda_1\mathbf{I} + \lambda_2\mathrm{rep}_Q\left(H_L^T H_L\right)\right. \\ &\qquad \left. + \rho\mathrm{diag}(\mathrm{vec}(\mathbf{E}_1\odot\mathbf{E}_1))\times\mathrm{vec}(\boldsymbol{\Phi})\right) \\ &\quad = \lambda_1\mathbf{1} + \rho\mathrm{vec}((\mathbf{E}_1\odot\mathbf{B} + \mathbf{E}_1\odot\mathbf{V})).\end{aligned} \tag{47}$$

The left-hand side of (47) is very sparse and, with this problem, can be efficiently solved [72] to obtain

$$\begin{aligned}&\boldsymbol{\Phi}^* \\ &= \mathrm{invec}\left(\lambda_1 1 + \boldsymbol{\rho}\mathrm{vec}((\mathbf{E}_1\odot\mathbf{B} + \mathbf{E}_1\odot\mathbf{V}))\,/\right. \\ &\qquad \left.\left(\lambda_1\mathbf{I} + \lambda_2\mathrm{rep}_Q\left(H_L^T H_L\right) + \boldsymbol{\rho}\mathrm{diag}(\mathrm{vec}(\mathbf{E}_1\odot\mathbf{E}_1))\right)\right)\end{aligned} \tag{48}$$

where symbol invec$(\cdot)$ is the inverse of the vectorization operator.

C. Dual Variables Update

The update for dual variables is given as follows:

$$\mathbf{V} \leftarrow \mathbf{V} + \mathbf{B} - \boldsymbol{\Phi}\odot\mathbf{E}_1. \tag{49}$$

Appendix C
Optimization of the ADMM for the T Subproblem

In this section, we describe the optimization of the scaling augmentation Lagrangian function (28) with respect to the variables of the temporal variation information.

A. Optimizing With Respect to $\mathcal{T}$

The optimization subproblem is expressed as follows:

$$\begin{aligned}\mathcal{T} &\in \arg\min_{\mathcal{B}} \mathcal{L}(\mathcal{T}, \mathcal{B}, \mathcal{V}) \\ &= \arg\min_{\mathcal{T}} \frac{1}{2}\|\mathbf{RT} - \mathbf{D}\|_2^2 \\ &\quad + \frac{\eta}{2}\|\mathbf{T} - ((\Phi\odot\mathbf{E}_1)\mathbf{A}_2 - \mathbf{E}_1\mathbf{A}_1)\|_2^2 \\ &\quad + \frac{\boldsymbol{\rho}}{2}\left(\|\mathcal{B} - \mathcal{T} + \mathcal{V}\|_2^2\right).\end{aligned} \tag{50}$$

By unfolding the third-order tensor $\mathcal{T}$, $\mathcal{B}$, and $\mathcal{V}$ into a matrix, we can rewrite the equation as

$$\begin{aligned}\mathcal{T} &\in \arg\min_{\mathcal{B}} \mathcal{L}(\mathcal{T}, \mathcal{B}, \mathcal{V}) \\ &= \arg\min_{\mathcal{T}} \frac{1}{2}\|\mathbf{RT} - \mathbf{D}\|_F^2 \\ &\quad + \frac{\eta}{2}\|\mathbf{T} - ((\boldsymbol{\Phi}\odot\mathbf{E}_1)\mathbf{A}_2 - \mathbf{E}_1\mathbf{A}_1)\|_F^2 \\ &\quad + \frac{\boldsymbol{\rho}}{2}\left(\|\mathbf{B} - \mathbf{T} + \mathbf{V}\|_F^2\right).\end{aligned} \tag{51}$$

Let $\partial\boldsymbol{\mathcal{L}}/\partial\mathbf{T}$ equal to zero, and we can obtain

$$\begin{aligned}&\left(\frac{1}{\eta+\rho}\mathbf{R}^T\mathbf{R} + 1\right)\mathbf{T} \\ &\quad = \frac{1}{\eta+\rho}\left(\mathbf{R}^T\mathbf{D} + \eta((\boldsymbol{\Phi}\odot\mathbf{E}_1)\mathbf{A}_2 - \mathbf{E}_1\mathbf{A}_1) + \rho(\mathbf{B} + \mathbf{V})\right).\end{aligned} \tag{52}$$

The solution is given as follows:

$$\begin{aligned}T^* &= \left(\frac{1}{\eta+\rho}\mathbf{R}^T\mathbf{R} + 1\right)^{-1} \\ &\quad \times\left(\frac{1}{\eta+\rho}\left(\mathbf{R}^T\mathbf{D} + \eta((\Phi\odot\mathbf{E}_1)\mathbf{A}_2 - \mathbf{E}_1\mathbf{A}_1) + \rho(\mathbf{B} + \mathbf{V})\right)\right).\end{aligned} \tag{53}$$

B. Optimizing With Respect to $\mathcal{B}$

The optimization subproblem on variable $\mathcal{B}$ can be formulated as

$$\begin{aligned}\mathcal{B} &\in \arg\min_{\mathcal{B}} \mathcal{L}(\mathcal{T}, \mathcal{B}, \mathcal{V}) \\ &= \arg\min_{\mathcal{B}} \frac{\rho}{2}\left(\|\mathcal{B} - \mathcal{T} + \mathcal{V}\|_2^2\right) + \lambda_T\sum_{k=1}^{K}\left\|\mathcal{B}^k\right\|_{\mathrm{TMR}}.\end{aligned} \tag{54}$$

We can solve the problem of (54) for each cluster independently

$$\arg\min_{\mathcal{B}^k} \frac{\rho}{2}\left(\left\|\mathcal{B}^k - \mathcal{T}^k + \mathcal{V}^k\right\|_2^2\right) + \lambda_T\sum_{k=1}^{K}\left\|\mathcal{B}^k\right\|_{\mathrm{TMR}} \tag{55}$$

where $\mathcal{B}^k$ and $\mathcal{V}^k$ are the tensor formed by the kth cluster grouping patch in $\mathcal{B}$ and $\mathcal{V}$, respectively.

According to the definition of tensor multirank $\|\mathcal{A}\|_{\mathrm{TMR}}$ in (18), the problem (55) can be rewritten as

$$\arg\min_{\bar{\mathcal{B}}^k} \frac{\rho}{2q}\left(\left\|\bar{\mathcal{B}}^k - \bar{\mathcal{G}}^k\right\|_2^2\right) + \lambda_T\sum_{i=1}^{q} LS\left\|\bar{\mathcal{B}}^k(:,:,i)\right\|_{\mathrm{TMR}} \tag{56}$$

where $\bar{\mathcal{B}}^k = \mathcal{F}(\mathcal{B}^k, [], 3)$ and $\bar{\mathcal{G}}^k = \mathcal{F}(\mathcal{T}^k - \mathcal{V}^k, [], 3)$ are obtained by performing Fourier transform on tensor $\mathcal{B}^k$ and $\mathcal{T}^k - \mathcal{V}^k$ along the third dimension, respectively. The estimation of the tensor $\bar{\mathcal{B}}^k$ is obtained by computing each front slice.

For the LS problem in (56), it has been proved in [1] that there are closed-form solutions with

$$\bar{\mathcal{B}}^k(:,:,i) = \mathbf{U}\hat{\boldsymbol{\Sigma}}V \tag{57}$$

where the matrices $\mathbf{U}$ and $\mathbf{V}$ are obtained by performing SVD on $\bar{\mathcal{G}}^k(:,:,i)$, with $\bar{\mathcal{G}}^k(:,:,i) = \mathbf{U}\boldsymbol{\Sigma}V$,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:25:27 UTC from IEEE Xplore. Restrictions apply.

$\hat{\mathbf{\Sigma}} = \text{diag}(\mathrm{S}_{\lambda_\mathbf{T}/\boldsymbol{\rho},\varepsilon}(\sigma_1), \mathrm{S}_{\lambda_\mathbf{T}/\boldsymbol{\rho},\varepsilon}(\sigma_2), \ldots, \mathrm{S}_{\lambda_\mathbf{T}/\boldsymbol{\rho},\varepsilon}(\sigma_\mathrm{n}))$, σ_i defines the ith singular of $\mathbf{\Sigma}$, and $\mathrm{S}_{\lambda_\mathbf{T}/\boldsymbol{\rho},\varepsilon}(\sigma)$ is the threshold operator, given by

$$\mathrm{S}_{\frac{\lambda_\mathbf{T}}{\rho},\varepsilon}(x) = \begin{cases} 0, & \text{if } c_\mathbf{2} \le 0 \\ \text{sign}(x)\left(\dfrac{c_\mathbf{1} + \sqrt{c_\mathbf{2}}}{2}\right), & \text{if } c_\mathbf{2} > 0 \end{cases} \tag{58}$$

where $0 < \varepsilon < \min(\sqrt{\lambda_\mathbf{T}}, (\lambda_\mathbf{T}/\sigma_1))$, $c_1 = |x| - \varepsilon$, and $c_2 = (c_1)^2 - 4(b - \varepsilon|x|)$. After obtaining $\bar{\mathcal{B}}^k$, $\mathcal{B}^k$ can be acquired by performing inverse Fourier transform on $\bar{\mathcal{B}}^k$. Finally, all clusters of $\mathcal{B}^k$ are aggregated to get $\mathcal{B}$.

C. Dual Variables Update

The update for dual variables is given as follows:

$$\mathcal{V} \leftarrow \mathcal{V} + \mathcal{B} - \mathcal{T}. \tag{59}$$

REFERENCES

[1] E. Adam, O. Mutanga, and D. Rugege, "Multispectral and hyperspectral remote sensing for identification and mapping of wetland vegetation: A review," *Wetlands Ecology Manage.*, vol. 18, no. 3, pp. 281–296, Jun. 2010.

[2] M. Wójtowicz, A. Wójtowicz, and J. Piekarczyk, "Application of remote sensing methods in agriculture," *Commun. Biometry Crop Sci.*, vol. 11, no. 1, pp. 31–50, 2016.

[3] K. Liu et al., "Mapping coastal wetlands using transformer in transformer deep network on China ZY1-02D hyperspectral satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3891–3903, 2022.

[4] K. He et al., "A dual global–local attention network for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5527613.

[5] Y. Yuan et al., "Multi-resolution collaborative fusion of SAR, multispectral and hyperspectral images for coastal wetlands mapping," *Remote Sens.*, vol. 14, no. 14, p. 3492, Jul. 2022.

[6] L. Xie, X. Feng, C. Zhang, Y. Dong, J. Huang, and J. Cheng, "A framework for soil salinity monitoring in coastal wetland reclamation areas based on combined unmanned aerial vehicle (UAV) data and satellite data," *Drones*, vol. 6, no. 9, p. 257, Sep. 2022.

[7] R. Dian, T. Shan, W. He, and H. Liu, "Spectral super-resolution via model-guided cross-fusion network," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jan. 27, 2023, doi: 10.1109/TNNLS.2023.3238506.

[8] M. Dalla Mura, S. Prasad, F. Pacifici, P. Gamba, J. Chanussot, and J. A. Benediktsson, "Challenges and opportunities of multimodality and data fusion in remote sensing," *Proc. IEEE*, vol. 103, no. 9, pp. 1585–1601, Sep. 2015.

[9] P. Ghamisi et al., "Multisource and multitemporal data fusion in remote sensing: A comprehensive review of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 1, pp. 6–39, Mar. 2019.

[10] Q. Liu, X. Chen, X. Meng, H. Chen, F. Shao, and W. Sun, "Dual-task interactive learning for unsupervised spatio-temporal–spectral fusion of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5402015.

[11] H. Shen, X. Meng, and L. Zhang, "An integrated framework for the spatio-temporal–spectral fusion of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7135–7148, Dec. 2016.

[12] P. J. Zarco-Tejada, S. L. Ustin, and M. L. Whiting, "Temporal and spatial relationships between within-field yield variability in cotton and high-spatial hyperspectral remote sensing imagery," *Agronomy J.*, vol. 97, no. 3, pp. 641–653, May 2005.

[13] L. Loncan et al., "Hyperspectral pansharpening: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 3, no. 3, pp. 27–46, Sep. 2015.

[14] N. Yokoya, C. Grohnfeldt, and J. Chanussot, "Hyperspectral and multispectral data fusion: A comparative review of the recent literature," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 2, pp. 29–56, Jun. 2017.

[15] X. Meng et al., "A large-scale benchmark data set for evaluating pansharpening performance: Overview and implementation," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 1, pp. 18–52, Mar. 2021.

[16] R. Dian, S. Li, and X. Kang, "Regularizing hyperspectral and multispectral image fusion by CNN denoiser," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 3, pp. 1124–1135, Mar. 2021.

[17] R. Dian, A. Guo, and S. Li, "Zero-shot hyperspectral sharpening," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 10, pp. 12650–12666, Oct. 2023.

[18] P. Wang, H. Yao, C. Li, G. Zhang, and H. Leung, "Multiresolution analysis based on dual-scale regression for pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5406319.

[19] C. Thomas, T. Ranchin, L. Wald, and J. Chanussot, "Synthesis of multispectral images to high spatial resolution: A critical review of fusion methods based on remote sensing physics," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 5, pp. 1301–1312, May 2008.

[20] P. Chavez, S. C. Sides, and J. A. Anderson, "Comparison of three different methods to merge multiresolution and multispectral data-Landsat TM and SPOT panchromatic," *Photogramm. Eng. Remote Sens.*, vol. 57, no. 3, pp. 295–303, 1991.

[21] C. A. Laben and B. V. Brower, "Process for enhancing the spatial resolution of multispectral imagery using pan-sharpening," U.S. Patent, US6 011 875 A, Apr. 4, 2000.

[22] G. Vivone, R. Restaino, M. D. Mura, G. Licciardi, and J. Chanussot, "Contrast and error-based fusion schemes for multispectral image pansharpening," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 5, pp. 930–934, May 2014.

[23] B. Aiazzi, L. Alparone, S. Baronti, A. Garzelli, and M. Selva, "MTF-tailored multiscale fusion of high-resolution MS and pan imagery," *Photogrammetric Eng. Remote Sens.*, vol. 72, no. 5, pp. 591–596, May 2006.

[24] K. He, J. Sun, and X. Tang, "Guided image filtering," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 6, pp. 1397–1409, Jun. 2013.

[25] L. Alparone, S. Baronti, B. Aiazzi, and A. Garzelli, "Spatial methods for multispectral pansharpening: Multiresolution analysis demystified," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 5, pp. 2563–2576, May 2016.

[26] X. Han, J. Yu, J.-H. Xue, and W. Sun, "Hyperspectral and multispectral image fusion using optimized twin dictionaries," *IEEE Trans. Image Process.*, vol. 29, pp. 4709–4720, 2020.

[27] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2012.

[28] X.-H. Han, B. Shi, and Y. Zheng, "Self-similarity constrained sparse representation for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 27, no. 11, pp. 5625–5637, Nov. 2018.

[29] N. Akhtar, F. Shafait, and A. Mian, "Sparse spatio-spectral representation for hyperspectral image super-resolution," in *Proc. Eur. Conf. Comput. Vis.*, Zurich, Switzerland, Sep. 2014, pp. 63–78.

[30] R. Dian, S. Li, L. Fang, T. Lu, and J. M. Bioucas-Dias, "Nonlocal sparse tensor factorization for semiblind hyperspectral and multispectral image fusion," *IEEE Trans. Cybern.*, vol. 50, no. 10, pp. 4469–4480, Oct. 2020.

[31] Y. Wang, X. Chen, Z. Han, and S. He, "Hyperspectral image super-resolution via nonlocal low-rank tensor approximation and total variation regularization," *Remote Sens.*, vol. 9, no. 12, p. 1286, Dec. 2017.

[32] M. Simoes, J. Bioucas-Dias, L. B. Almeida, and J. Chanussot, "A convex formulation for hyperspectral image superresolution via subspace-based regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3373–3388, Jun. 2015.

[33] K. Zhang, M. Wang, and S. Yang, "Multispectral and hyperspectral image fusion based on group spectral embedding and low-rank factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1363–1371, Mar. 2017.

[34] Y. Zhou, L. Feng, C. Hou, and S.-Y. Kung, "Hyperspectral and multispectral image fusion based on local low rank and coupled spectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5997–6009, Oct. 2017.

[35] T. Gelvez-Barrera, H. Arguello, and A. Foi, "Joint nonlocal, spectral, and similarity low-rank priors for hyperspectral–multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5537112.

[36] S. Li, R. Dian, L. Fang, and J. M. Bioucas-Dias, "Fusing hyperspectral and multispectral images via coupled sparse tensor factorization," *IEEE Trans. Image Process.*, vol. 27, no. 8, pp. 4118–4130, Aug. 2018.

[37] R. Dian, S. Li, and L. Fang, "Learning a low tensor-train rank representation for hyperspectral image super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 9, pp. 2672–2683, Sep. 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:25:27 UTC from IEEE Xplore. Restrictions apply.

[38] R. Dian and S. Li, "Hyperspectral image super-resolution via subspace-based low tensor multi-rank regularization," *IEEE Trans. Image Process.*, vol. 28, no. 10, pp. 5135–5146, Oct. 2019.

[39] Y. Xu, Z. Wu, J. Chanussot, and Z. Wei, "Nonlocal patch tensor sparse representation for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 28, no. 6, pp. 3034–3047, Jun. 2019.

[40] Y. Xu, Z. Wu, J. Chanussot, P. Comon, and Z. Wei, "Nonlocal coupled tensor CP decomposition for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 348–362, Jan. 2020.

[41] Y. Xu, Z. Wu, J. Chanussot, and Z. Wei, "Hyperspectral images super-resolution via learning high-order coupled tensor ring representation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 11, pp. 4747–4760, Nov. 2020.

[42] X. Han, J. Yu, J. Luo, and W. Sun, "Hyperspectral and multispectral image fusion using cluster-based multi-branch BP neural networks," *Remote Sens.*, vol. 11, no. 10, p. 1173, May 2019.

[43] R. Dian, S. Li, A. Guo, and L. Fang, "Deep hyperspectral image sharpening," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 11, pp. 5345–5355, Nov. 2018.

[44] Q. Xie, M. Zhou, Q. Zhao, D. Meng, W. Zuo, and Z. Xu, "Multispectral and hyperspectral image fusion by MS/HS fusion net," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 1585–1594.

[45] Y. Qu, H. Qi, and C. Kwan, "Unsupervised sparse Dirichlet-net for hyperspectral image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2511–2520.

[46] R. Dian, S. Li, B. Sun, and A. Guo, "Recent advances and new guidelines on hyperspectral and multispectral image fusion," *Inf. Fusion*, vol. 69, pp. 40–51, May 2021.

[47] X. Zhu, F. Cai, J. Tian, and T. Williams, "Spatiotemporal fusion of multisource remote sensing data: Literature survey, taxonomy, principles, applications, and future directions," *Remote Sens.*, vol. 10, no. 4, p. 527, Mar. 2018.

[48] Q. Wang and P. M. Atkinson, "Spatio-temporal fusion for daily Sentinel-2 images," *Remote Sens. Environ.*, vol. 204, pp. 31–42, Jan. 2018.

[49] Q. Wang, X. Ding, X. Tong, and P. M. Atkinson, "Spatio-temporal spectral unmixing of time-series images," *Remote Sens. Environ.*, vol. 259, Jun. 2021, Art. no. 112407.

[50] Q. Wang, Y. Tang, X. Tong, and P. M. Atkinson, "Virtual image pair-based spatio-temporal fusion," *Remote Sens. Environ.*, vol. 249, Nov. 2020, Art. no. 112009.

[51] F. Gao, J. Masek, M. Schwaller, and F. Hall, "On the blending of the Landsat and MODIS surface reflectance: Predicting daily Landsat surface reflectance," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 8, pp. 2207–2218, Aug. 2006.

[52] X. Zhu, J. Chen, F. Gao, X. Chen, and J. G. Masek, "An enhanced spatial and temporal adaptive reflectance fusion model for complex heterogeneous regions," *Remote Sens. Environ.*, vol. 114, no. 11, pp. 2610–2623, Nov. 2010.

[53] T. Hilker et al., "A new data fusion model for high spatial- and temporal-resolution mapping of forest disturbance based on Landsat and MODIS," *Remote Sens. Environ.*, vol. 113, no. 8, pp. 1613–1627, Aug. 2009.

[54] J. Amorós-López et al., "Multitemporal fusion of Landsat/TM and ENVISAT/MERIS for crop monitoring," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 23, pp. 132–141, Aug. 2013.

[55] R. Zurita-Milla, G. Kaiser, J. G. P. W. Clevers, W. Schneider, and M. E. Schaepman, "Downscaling time series of MERIS full resolution data to monitor vegetation seasonal dynamics," *Remote Sens. Environ.*, vol. 113, no. 9, pp. 1874–1885, Sep. 2009.

[56] C. M. Gevaert and F. J. García-Haro, "A comparison of STARFM and an unmixing-based algorithm for Landsat and MODIS data fusion," *Remote Sens. Environ.*, vol. 156, pp. 34–44, Jan. 2015.

[57] B. Huang and H. Song, "Spatiotemporal reflectance fusion via sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 10, pp. 3707–3716, Oct. 2012.

[58] H. Song and B. Huang, "Spatiotemporal satellite image fusion through one-pair image learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 4, pp. 1883–1896, Apr. 2013.

[59] X. Liu, C. Deng, J. Chanussot, D. Hong, and B. Zhao, "StfNet: A two-stream convolutional neural network for spatiotemporal image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6552–6564, Sep. 2019.

[60] H. Song, Q. Liu, G. Wang, R. Hang, and B. Huang, "Spatiotemporal satellite image fusion using deep convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 3, pp. 821–829, Mar. 2018.

[61] Z. Tan, M. Gao, X. Li, and L. Jiang, "A flexible reference-insensitive spatiotemporal fusion model for remote sensing images using conditional generative adversarial network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5601413.

[62] X. Meng, Q. Liu, F. Shao, and S. Li, "Spatio-temporal–spectral collaborative learning for spatio-temporal fusion with land cover changes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5704116.

[63] L. Zhang, D. Fu, X. Sun, H. Chen, and X. She, "A spatial–temporal-spectral blending model using satellite images," *IOP Conf. Ser., Earth Environ. Sci.*, vol. 34, Apr. 2016, Art. no. 012042.

[64] Y. Zhao and B. Huang, "A hybrid image fusion model for generating high spatial–temporal-spectral resolution data using OLI-MODIS-Hyperion satellite imagery," *Int. J. Geological Environ. Eng.*, vol. 11, no. 9, pp. 869–874, 2017.

[65] Y. Tang, Q. Wang, X. Tong, and P. M. Atkinson, "Integrating spatio-temporal-spectral information for downscaling Sentinel-3 OLCI images," *ISPRS J. Photogramm. Remote Sens.*, vol. 180, pp. 130–150, Oct. 2021.

[66] H. Shen, "Integrated fusion method for multiple temporal-spatial–spectral images," in *Proc. 22nd Congr. (ISPRS)*, 2012, pp. 407–410.

[67] B. Huang, H. Zhang, H. Song, J. Wang, and C. Song, "Unified fusion of remote-sensing imagery: Generating simultaneously high-resolution synthetic spatial–temporal–spectral Earth observations," *Remote Sens. Lett.*, vol. 4, no. 6, pp. 561–569, Jun. 2013.

[68] C. Zhao, X. Gao, W. J. Emery, Y. Wang, and J. Li, "An integrated spatio-spectral–temporal sparse representation method for fusing remote-sensing images with different resolutions," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3358–3370, Jun. 2018.

[69] Y. Peng, W. Li, X. Luo, J. Du, Y. Gan, and X. Gao, "Integrated fusion framework based on semicoupled sparse tensor factorization for spatio-temporal–spectral fusion of remote sensing images," *Inf. Fusion*, vol. 65, pp. 21–36, Jan. 2021.

[70] W. Li, Y. Wang, N. Liu, C. Xiao, Z. Sun, and Q. Du, "Integrated spatio-spectral–temporal fusion via anisotropic sparsity constrained low-rank tensor approximation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5517416.

[71] J. Zhou, W. Sun, X. Meng, G. Yang, K. Ren, and J. Peng, "Generalized linear spectral mixing model for spatial–temporal–spectral fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5533216.

[72] T. Imbiriba, R. A. Borsoi, and J. C. Moreira Bermudez, "Generalized linear mixing model accounting for endmember variability," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2018, pp. 1862–1866.

[73] R. A. Borsoi, T. Imbiriba, and J. C. M. Bermudez, "Super-resolution for hyperspectral and multispectral image fusion accounting for seasonal spectral variability," *IEEE Trans. Image Process.*, vol. 29, pp. 116–127, 2020.

[74] Z. Zhang, G. Ely, S. Aeron, N. Hao, and M. Kilmer, "Novel methods for multilinear data completion and de-noising based on tensor-SVD," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 3842–3849.

[75] Q. Xie et al., "Multispectral images denoising by intrinsic tensor sparsity regularization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1692–1700.

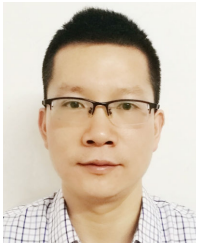
[76] X. Sun, L. Zhang, H. Yang, T. Wu, Y. Cen, and Y. Guo, "Enhancement of spectral resolution for remotely sensed multispectral image," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 5, pp. 2198–2211, May 2015.

[77] J. Wu, J. Aeschbacher, and R. Timofte, "In defense of shallow learned spectral reconstruction from RGB images," in *Proc. IEEE Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2017, pp. 471–479.

[78] L. Gao, D. Hong, J. Yao, B. Zhang, P. Gamba, and J. Chanussot, "Spectral superresolution of multispectral imagery with joint sparse and low-rank learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2269–2280, Mar. 2021.

[79] L. Drumetz, M.-A. Veganzones, S. Henrot, R. Phlypo, J. Chanussot, and C. Jutten, "Blind hyperspectral unmixing using an extended linear mixing model to address spectral variability," *IEEE Trans. Image Process.*, vol. 25, no. 8, pp. 3890–3905, Aug. 2016.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:25:27 UTC from IEEE Xplore. Restrictions apply.



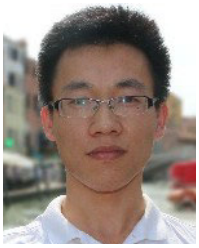
Weiwei Sun (Senior Member, IEEE) received the B.S. degree in surveying and mapping and the Ph.D. degree in cartography and geographic information engineering from Tongji University, Shanghai, China, in 2007 and 2013, respectively.

From 2011 to 2012, he studied at the Department of Applied Mathematics, University of Maryland, College Park, MD, USA, where he worked as a Visiting Scholar with the famous professor John Benedetto to study the dimensionality reduction of hyperspectral image. From 2014 to 2016, he studied at the State Key Laboratory for Information Engineering in Surveying, Mapping, and Remote Sensing (LIESMARS), Wuhan University, Wuhan, China, where he worked as a Post-Doctoral Researcher to study intelligent processing in hyperspectral imagery. From 2017 to 2018, he was a Visiting Scholar in hyperspectral image processing with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA. He is currently a Full Professor with Ningbo University, Ningbo, China. He has published more than 80 journal articles. His research interest includes hyperspectral image processing with machine learning.



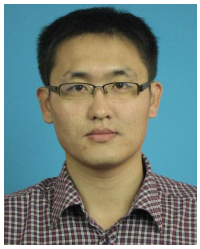
Jun Zhou received the B.S. degree in geomatic from Yunnan Normal University, Kunming, China, in 2019. He is currently pursuing the M.E degree with the School of Civil and Environmental Engineering and Geography Science, Ningbo University, Ningbo, China.

His research interests include remote sensing data fusion and machine learning techniques for remote sensing image processing.



Xiangchao Meng (Member, IEEE) received the B.S. degree in geographic information systems from the Shandong University of Science and Technology, Qingdao, China, in 2012, and the Ph.D. degree in cartography and geography information system from Wuhan University, Wuhan, China, in 2017.

He is currently an Associate Professor with the Faculty of Electrical Engineering and Computer Science, Ningbo University, Ningbo, China. His research interests include remote sensing image fusion and quality evaluation.



Gang Yang (Member, IEEE) received the M.S. degree in geographical information system from the Hunan University of Science and Technology, Xiangtan, China, in 2012, and the Ph.D. degree from the School of Resource and Environmental Sciences, Wuhan University, Wuhan, China, in 2016.

He is currently a Lecturer with Ningbo University, Ningbo, China. His research interests focus on missing information reconstruction of remote sensing images, cloud removal of remote sensing images, and remote sensing time-series products' temporal reconstruction.



Kai Ren received the B.S. degree in humanistic geography and urban–rural planning from Shaanxi Normal University, Xi'an, China, in 2018, and the M.S. degree in humanistic geography from Ningbo University, Ningbo, China, in 2021, where he is currently pursuing the Ph.D. degree with School of Information Science and Engineering.

His research interests include remote sensing data fusion and data analysis.



Jiangtao Peng (Senior Member, IEEE) received the B.S. and M.S. degrees from Hubei University, Wuhan, China, in 2005 and 2008, respectively, and the Ph.D. degree from the Institute of Automation, Chinese Academy of Sciences, Beijing, China, in 2011.

He is currently a Professor with the Faculty of Mathematics and Statistics, Hubei University. His research interests include machine learning and hyperspectral image processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:25:27 UTC from IEEE Xplore. Restrictions apply.